

CHRONOS

Calibrated Harmonic Representation Of Norm-deviations in Observed Scenes

A Trajectory-Based, Statistically Guaranteed Architecture for
Semantic Retrieval and Multi-Temporal Change Analysis of Satellite Imagery

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Abstract

Earth-observation archives have outgrown metadata catalogues. Analysts must already know *where* and *when* to look before they can look at all. Problem Statement 26227 asks for a system that makes such an archive queryable by *semantic content* and by *change over time*, operating fully on-premises, ingesting new acquisitions incrementally, preserving geospatial provenance, and — critically — suppressing false change arising from season, atmosphere, viewing geometry, registration error and sensor differences.

We observe that essentially every deployed and published system in this space commits the same foundational error: it represents a ground location as a *point* in an embedding space, and defines change as the *distance between two points*. That single modelling choice is the direct cause of the field’s three acknowledged open problems — seasonality is formally indistinguishable from change, detection thresholds are arbitrary, and semantic retrieval and change analysis end up as two disconnected subsystems.

CHRONOS adopts the opposite premise. A ground location is a *trajectory*: a curve in embedding space, sampled irregularly and unevenly by cloud, orbit and sensor availability. We fit a robust, quality-weighted harmonic-trend model to that curve per tile, treating the fit as a *tile fixed effect* that absorbs phenology; we then subtract a *time fixed effect* estimated from a phenologically matched control cohort, yielding a residual that is formally a difference-in-differences estimand of real change. We convert that residual into a decision via Mondrian split-conformal calibration and an anytime-valid e-process, obtaining a *distribution-free, finite-sample guarantee* on the false-alarm rate that holds under unbounded incremental ingestion without any multiple-testing correction over time. The same decomposition yields a season-invariant retrieval vector, a phenological fingerprint for unsupervised site discovery, and a change-evidence signal that composes with semantic similarity into a single ranking function. Relational language (“near a river”) — provably beyond CLIP-family encoders — is routed to a symbolic GIS operator rather than asked of the neural model.

This paper states the problem formally, analyses the four mandated data sources and exactly what they can and cannot support, develops the theory with proofs, specifies the complete system across thirteen implementation stages, gives complexity and storage analysis with worked numbers for a 200×200 km area of interest over three years, defines the evaluation protocol, and maps every deliverable back to the originating requirement.

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1 Introduction

1.1 Operational context

The problem originates with the Directorate General of Information Systems, Indian Army. That provenance determines several non-negotiable properties of any acceptable solution, and it is worth stating them before any technical discussion:

- (i) **Sovereignty is a hard constraint, not a preference.** The demonstration runs with network access disabled. No cloud service, no external API, no runtime model download, no remote basemap tile server. Everything is staged in advance and runs on local iron.
- (ii) **Precision dominates recall.** The problem statement says so explicitly. An analyst review queue that is 60% noise is operationally worthless regardless of how high its recall is; it consumes the scarcest resource in the loop, which is trained human attention.
- (iii) **Provenance is evidentiary.** Outputs must be defensible after the fact. Source scene, processing lineage, model version, algorithm parameters and analyst decisions must all survive into the export.
- (iv) **Evaluation is adversarial and held out.** Retrieval is scored against unseen semantic queries with relevance judgements; change is scored against an unseen labelled set containing both change *and no-change* cases. The no-change cases exist precisely to punish systems that confuse season with construction.

1.2 The core technical observation

Consider the standard pipeline, used by every embedding-based Earth-observation system we surveyed: tile the archive, encode each tile with a pretrained foundation model into a single vector, index the vectors for approximate nearest-neighbour search, and detect change by taking the distance between the vectors of two dates.

Write this formally. A tile i observed at times $t_1 < t_2$ yields $\mathbf{e}_i(t_1), \mathbf{e}_i(t_2) \in \mathbb{R}^d$, and change is declared when $\|\mathbf{e}_i(t_2) - \mathbf{e}_i(t_1)\| > \theta$ for some threshold θ . Now decompose the observed embedding into its true generative parts:

$$\mathbf{e}_i(t) = \underbrace{\boldsymbol{\alpha}_i}_{\text{what this place is}} + \underbrace{f_i(t)}_{\text{its seasonal cycle}} + \underbrace{\boldsymbol{\tau}(t)}_{\text{atmosphere, illumination, sensor state}} + \underbrace{\mathbf{c}_i(t)}_{\text{real change}} + \underbrace{\boldsymbol{\eta}_i(t)}_{\text{noise}}.$$

The bi-temporal difference is

$$\mathbf{e}_i(t_2) - \mathbf{e}_i(t_1) = [f_i(t_2) - f_i(t_1)] + [\boldsymbol{\tau}(t_2) - \boldsymbol{\tau}(t_1)] + [\mathbf{c}_i(t_2) - \mathbf{c}_i(t_1)] + \Delta\boldsymbol{\eta}.$$

The quantity of interest, $\Delta\mathbf{c}_i$, is one of four additive terms and is *not identified*. The phenological term Δf_i is frequently the largest of the four: the embedding distance between a rice paddy in April and the same paddy in August routinely exceeds the distance between a field in April and a construction site in April. No threshold θ can separate them, because the information required to do so was destroyed when the trajectory was collapsed to two points.

Every heuristic in the false-alarm-suppression literature — pseudo-invariant-feature radiometric normalisation, IR-MAD, seasonal-invariant contrastive pretraining, style-transfer normalisation, frequency-domain filtering — is an attempt to attack one of the nuisance terms after the information needed to attack it properly has already been thrown away.

Thesis. Do not collapse the trajectory. Fit the nuisance structure explicitly on the full time series, remove it by construction, and calibrate what remains. Retrieval, change detection, change dating, clustering and false-alarm suppression then all become operations on one decomposed representation, rather than five separate systems held together by thresholds.

1.3 Contributions

- C1. Harmonic Embedding Fields.** A robust, quality-weighted harmonic–trend–break decomposition fitted to full foundation-model embedding trajectories, generalising CCDC [1] from seven bands of surface reflectance to a d -dimensional learned semantic space. Closed-form, ridge-regularised, and updatable by recursive least squares. (§5.1, Stage 6.)
- C2. Change as a difference-in-differences estimand.** A two-way fixed-effects formulation in which the harmonic fit is the tile effect and a phenologically matched control cohort supplies the time effect. We state the identification assumptions explicitly and show that matching on harmonic coefficients is matching on the confounder. (§5.3, Stage 7.)
- C3. Conformal change control.** Mondrian split-conformal calibration on label-free null sets harvested from stable segments, giving a distribution-free finite-sample bound on the false-alarm rate, plus Benjamini–Yekutieli control of the false discovery rate across the analyst’s review queue. The circularity inherent in self-harvested nulls is bounded rather than ignored: Lemma 11 shows contamination biases the threshold *upward*, so validity is preserved and the cost is paid in power. (§5.4, Stage 7.)
- C4. Anytime-valid sequential detection.** A conformal *test martingale* over smoothed, out-of-fold conformal p-values, whose validity is preserved under unbounded incremental ingestion by Ville’s inequality — the correct statistical frame for an archive that never stops growing. We are explicit that the naive e-process fails here, because marginal conformal validity does not imply the conditional validity a supermartingale argument needs, and we construct the detector so the required exchangeability actually holds. (§5.5, Stage 7.)
- C5. Mode-independence of the guarantee.** The statistical layer does not depend on the harmonic layer. Where the archive is too short for a seasonal model — or where the evaluation supplies bare image pairs — the system degrades to a designed bi-temporal mode in which C2 and C3 operate unchanged. (§5.2.)
- C6. Observability-bounded change dating.** The reported answer to “when did this change” is an interval bounded by the last clean pre-break observation and the first confirmed post-break observation, with the intervening unusable acquisitions enumerated as an explicit observability gap. (§5.6, Stage 7.)
- C7. Neuro-symbolic query compilation.** A typed query algebra that routes semantic predicates to embedding search, temporal predicates to the change engine, and *relational spatial predicates* to *PostGIS* — rather than asking a CLIP-family encoder to do something it is documented to fail at [18, 19]. (§5.9.2, Stage 9.)
- C8. Decomposed tri-vector index with late-interaction reranking.** Three vectors per tile (season-invariant semantics, phenological fingerprint, sparse residual) instead of one, plus ColBERT-style MaxSim reranking of the top candidates for sub-tile localisation without persisting patch vectors. (§7, Stage 8.)
- C9. Closed-form incrementality throughout.** Every stateful component updates in closed form on arrival of a new acquisition: recursive least squares for the harmonic fit, Sherman–Morrison for the covariance, HNSW insertion for the index, multiplicative update for the e-process, adaptive conformal for the threshold. No component requires a rebuild. (§7, Stage 11.)
- C10. Feedback that tightens the guarantee.** Analyst rejections are exactly the null-hypothesis samples that conformal calibration requires. Human review therefore improves not merely the ranking but the statistical guarantee itself. (§7, Stage 10.)

2 Requirement Decomposition

We first restate the problem statement as testable engineering requirements. Every subsequent design decision is traced back to one of these. The traceability matrix is given in §11.

ID	Source	Requirement
R1.1	PS 2.2.1	Free-text natural-language search over imagery tiles.
R1.2	PS 2.2.1	Image-to-image search for visually and semantically similar locations.
R1.3	PS 2.2.1	Rank-ordered results.
R1.4	PS 2.2.1	Refinement by AOI, date range, sensor and other metadata filters.
R1.5	PS 2.2.1	Support compound/relational queries (“newly built structures near a river”).
R2.1	PS 2.2.2	For a given AOI and time window, identify appearance, disappearance, expansion and contraction of features.
R2.2	PS 2.2.2	Classify change type: construction, clearance, water-extent variation, road development.
R2.3	PS 2.2.2	Estimate the earliest available observation at which the change is supported by <i>usable</i> imagery.
R3.1	PS 2.2.3	Treat season, illumination, view angle, cloud, haze, snow, shadow, radiometric inconsistency and imperfect co-registration as confounders, not as change.
R3.2	PS 2.2.3	Employ quality masks, normalisation and confidence estimates.
R3.3	PS 2.2.3	Favour precision over indiscriminate recall.
R4.1	PS 2.2.4	Unsupervised / embedding-based grouping of similar sites across a wide area.
R4.2	PS 2.2.4	Query-by-example discovery: one site of interest → comparable sites, without hand-constructing a new query per site.
R5.1	PS 2.2.5	Ranked review queue with before/after evidence, location, acquisition time, sensor, confidence and processing history.
R5.2	PS 2.2.5	Analyst confirm/reject, persisted to an audit trail.
R5.3	PS 2.2.5	Feedback used for subsequent reranking or refinement.
R5.4	PS 2.2.5	Exports retain source-scene and processing provenance.
R6.1	PS 2.2.6	Efficient vector or equivalent indexing.
R6.2	PS 2.2.6	Incremental addition of new imagery <i>without a complete index rebuild</i> .
R6.3	PS 2.2.6	Complete on-premises operation; no cloud services or external APIs at evaluation.
R6.4	PS 2.2.6	Georeferencing and acquisition metadata preserved end to end.
R6.5	PS 2.2.6	Ingest GeoTIFF and Cloud Optimized GeoTIFF (COG).
R7.1	PS 2.2.7	Full demonstration with network access disabled after staging.
R7.2	PS 2.2.7	Pretrained public models permitted; origin and licence declared; weights packaged for offline use.

ID	Source	Requirement
R7.3	PS 2.2.7	Public or organiser-generated imagery only.
R8.1	PS 2.3	Deliver source code.
R8.2	PS 2.3	Deliver architecture note.
R8.3	PS 2.3	Deliver index-build and incremental-ingestion procedure.
R8.4	PS 2.3	Deliver model and dataset provenance.
R8.5	PS 2.3	Deliver reproducible evaluation report stating indexed area, number of scenes/tiles, build time, storage footprint, query latency and hardware.

Table 1: Requirements extracted from Problem Statement 26227.

A note on R1.5 and R2.3. These two requirements are, in our reading, the discriminating ones. R1.5 is stated only implicitly, through the worked example “newly built structures near a river”, but that single phrase contains a semantic predicate, a temporal predicate and a *relational spatial* predicate, and no published remote-sensing retrieval system handles the third. R2.3 — “the earliest available observation at which the change is supported by *usable* imagery” — is not the same question as “when did the change occur”. It is a question about the interaction between the change and the observation schedule, and answering it correctly requires propagating the quality mask into the temporal inference. Both are addressed directly by our design rather than approximated.

3 Data Analysis

The problem statement names four families of imagery. This section characterises each, then analyses what the union of them can and cannot support. Getting this right up front determines the tiling scheme, the temporal model order, the sensor-harmonisation strategy and, bluntly, which of the organisers’ example queries are answerable at all.

3.1 Copernicus Sentinel-2 (optical, multispectral)

- **Bands.** 13 spectral bands. Four at 10 m (B2 blue, B3 green, B4 red, B8 NIR); six at 20 m (B5, B6, B7 red-edge; B8A narrow NIR; B11, B12 SWIR); three at 60 m (B1 aerosol, B9 water vapour, B10 cirrus — B10 absent from L2A).
- **Products.** L1C (top-of-atmosphere reflectance) and L2A (bottom-of-atmosphere / surface reflectance, with the Scene Classification Layer, SCL). **We use L2A** because atmospheric correction removes a substantial fraction of the $\tau(t)$ nuisance term before it ever reaches our model, and because SCL provides a free first-pass quality mask.
- **Tiling.** 110 km MGRS tiles, UTM/WGS84, with overlap between adjacent tiles.
- **Revisit.** 5 days at the equator with the two-satellite constellation; better at higher latitudes due to swath overlap. Over India, an effective 2–5 day cadence.
- **Geolocation.** Specified 12.5 m CE@95.5%. *Multi-temporal misregistration measured empirically is 1.2 pixels between same-orbit acquisitions and up to 2.8 pixels between different-orbit acquisitions at 10 m [5].* That upper figure is 28 m — more than an entire small building — and it is the single most under-appreciated false-alarm source in this problem. Requirement R3.1 names it explicitly.

3.2 Copernicus Sentinel-1 (C-band SAR)

- **Constellation status (as of September 2026).** Replenishment is complete. Sentinel-1C and Sentinel-1D re-established the nominal six-day interferometric repeat cycle on 24 June 2026; Sentinel-1A operations were terminated on 29 June 2026 [39]. **Design implication:** the six-day revisit grid is phase-shifted by one day relative to the historical S1A/S1B timeline. Any pipeline with hard-coded acquisition dates or an assumed repeat lattice will silently mis-align across that boundary. Our harmonic model consumes irregular timestamps natively and is therefore immune, but the ingestion metadata must record the platform.
- **Mode and product.** Interferometric Wide (IW) swath, dual polarisation VV+VH over land. Ground Range Detected (GRD) at 10 m pixel spacing is the practical choice; Single Look Complex (SLC) is required only if interferometric coherence is used (see §10).
- **Why it matters here.** SAR is cloud- and illumination-independent. In the Indian monsoon belt it is frequently the *only* usable observation for months at a time. It is also physically complementary: construction produces double-bounce returns that raise σ_{VV}^0 ; open water is specular and collapses backscatter; vegetation removal reduces volume scattering and therefore σ_{VH}^0 .
- **Cost.** SAR requires its own preprocessing chain (orbit correction, border and thermal noise removal, radiometric calibration to σ^0 , terrain flattening to γ^0 , speckle filtering, layover/shadow masking from a DEM) and speckle is multiplicative, not additive, which changes the noise model.

3.3 USGS Landsat Collection 2

- **Sensors.** Landsat 8 OLI/TIRS and Landsat 9 OLI-2/TIRS-2; the archive extends back to Landsat 4/5 TM and 7 ETM+.
- **Resolution.** 30 m multispectral, 15 m panchromatic, 100 m thermal (resampled to 30 m).
- **Products.** Collection 2 Level-2 surface reflectance with the QA_PIXEL band (CFMask-derived cloud, shadow, snow, water flags).
- **Revisit.** 16 days per satellite; 8 days for the L8+L9 pair.
- **Value.** Depth. Landsat gives a multi-decadal baseline that Sentinel-2 (2015 onwards) cannot. For estimating a stable phenological prior at a location, or for establishing that a site was undeveloped for twenty years before a break, Landsat is irreplaceable.
- **Cost.** Its 30 m grid does not align with Sentinel-2’s 10 m grid, its geolocation accuracy is looser (≈ 35 m CE@95.5%), and combined L8→S2 co-registration error is roughly 38 m before correction [5].

3.4 NRSC / ISRO Bhuvan

This source requires care, because what is *visible* on Bhuvan and what is *freely downloadable* are different sets, and the difference matters for our design.

- **Freely downloadable** via the NRSC Open EO Data Archive (NOEDA): Resourcesat-1 AWiFS Ortho at 56 m, Resourcesat-1 LISS-III Ortho at 23.5–24 m (four bands: green, red, NIR, SWIR; 141 km swath; ortho-rectified to WGS84; GeoTIFF tiles), IMS-1 HySI at 500 m, and CartoDEM at 1 arc-second. Released under the National Data Sharing and Accessibility Policy (NDSAP).

- **Not freely downloadable:** LISS-IV at 5.8 m and Cartosat panchromatic are available for *visualisation* on the Bhuvan 2D/3D viewer but are not part of the open download archive; data at 1 m and finer is hosted for visualisation only. Archive ordering for these has largely migrated to the Bhoonidhi portal.
- **Latency.** Free seasonal AWiFS and LISS-III products carry roughly a two-year lag.
- **Design consequence.** CartoDEM is genuinely valuable and we will use it (terrain correction, SAR layover/shadow masking, illumination modelling, and as a stratification variable for the control cohort). LISS-III adds a useful independent optical sensor and a SWIR band. But **Bhuvan does not raise our effective ground sampling distance**, and any design predicated on 5.8 m Indian imagery being freely available is predicated on a false premise.

3.5 The resolution–semantics analysis

This subsection exists because it determines what we can honestly promise.

Let ρ denote ground sampling distance and L the characteristic length of a target. Reliable detection of an object as an *object* requires roughly $L/\rho \gtrsim 3$ –5 pixels across; reliable classification requires more.

Target	L (m)	L/ρ at 10 m	L/ρ at 30 m	Verdict
Reservoir, lake, river reach	10^2 – 10^4	$\gg 10$	$\gg 10$	Detectable and classifiable
Deforestation / clearance patch	50 – 10^3	5–100	2–33	Detectable
Large industrial building	50–200	5–20	1.7–6.7	Detectable at 10 m
Road (new alignment)	width 7–20	0.7–2	0.2–0.7	<i>Linear</i> detection only
Residential structure	8–15	0.8–1.5	0.3–0.5	Sub-pixel; aggregate only
Heavy truck	8–18	0.8–1.8	0.3–0.6	Not resolvable
Light vehicle	4–5	0.4–0.5	0.1–0.2	Not resolvable

Table 2: Target size versus available ground sampling distance across the mandated data sources.

Remark 1 (On the second worked query). The problem statement offers “large vehicle concentrations on open ground” as an example query. At the resolutions the mandated open data sources actually provide, an individual vehicle occupies well under one pixel. This query is *not answerable as literally stated* from Sentinel-2, Landsat, or the freely downloadable Bhuvan products.

We do not treat this as a reason to narrow the deliverable. We treat it as a design requirement, and we respond in three ways, all of which we consider more useful than silently returning plausible nonsense:

1. **Aggregate-signature retrieval.** A concentration of vehicles or materiel on open ground is not resolvable object-wise but *is* detectable as a change in aggregate surface properties: a bare or vegetated surface acquiring high-density metallic scatterers produces a characteristic rise in SAR backscatter (especially σ_{VV}^0 , from corner reflection) with little or no corresponding change in optical NDVI or NDBI. That optical/SAR *discordance* is itself a discriminative signature, and our multi-sensor residual vector captures it directly. We index and expose it.
2. **Honest confidence degradation.** The conformal layer reports calibrated confidence. For a query whose target is sub-pixel, confidences will be low and the system will say so, rather than returning a confidently ranked list of false positives. A system that knows what it cannot see is more valuable than one that does not.
3. **Resolution-aware capability declaration.** The architecture note will state per query class the minimum GSD required, and the UI will surface this. If the organisers supply higher resolution imagery, the same pipeline consumes it — our encoders and tiling are GSD-parameterised, not GSD-assuming.

3.6 Harmonisation problem statement

The union of these sources is heterogeneous along five axes simultaneously. Each is a false-alarm source under R3.1 and each requires an explicit treatment.

Axis	Heterogeneity	Stage
Geometric	10 / 20 / 30 / 56 m GSD; different UTM zones; multi-temporal misregistration to 2.8 px; cross-sensor error to 38 m	Stage 2
Radiometric	Different band centres and widths; different atmospheric correction; different calibration histories; BRDF/view-angle effects	Stage 3
Temporal	5-day (S2), 6-day (S1), 8/16-day (Landsat), seasonal (Bhuvan); irregular after cloud masking; two-year latency on Bhuvan free products	Stage 6
Modal	Optical reflectance versus SAR backscatter — physically different quantities with different noise models (additive vs. multiplicative speckle)	Stages 4, 5
Semantic	4 bands (LISS-III) to 13 (S2) to 2 polarisations (S1); encoders expect fixed band sets	Stage 5

Table 3: The five-axis harmonisation problem and where each axis is addressed.

Remark 2 (Do not over-correct). A standard instinct is to force everything onto one radiometric scale by aggressive normalisation. We deliberately do not. Perfect cross-sensor radiometric harmonisation is unattainable, and each correction injects its own error. Instead we harmonise geometrically (which *is* attainable to sub-pixel accuracy) and then *condition on sensor* in the Mondrian conformal stratification (§5.4). Residual cross-sensor bias is thereby absorbed into the calibration rather than propagated as an uncorrected error. This is a substantially more honest treatment and it is cheaper. It is one of the more consequential design decisions in this paper.

4 Formal Problem Statement

4.1 Notation

Symbol	Meaning
$\mathcal{A}$	The archive: a set of georeferenced scenes.
$\mathcal{S}$	Sensor set, $s \in \mathcal{S} = \{\text{S2, S1, L8/9, LISS-III}\}$.
$\mathcal{G}$	Tile grid; $N = \mathcal{G} $ tiles, indexed i .
g_i	Geometry (polygon, WGS84) of tile i .
$\mathcal{T}_i$	Set of acquisition times with a valid observation of tile i .
$X_i(t)$	Raster stack for tile i at time t , $\in \mathbb{R}^{B_s \times P \times P}$.
$q_i(t) \in [0, 1]$	Usability weight of that observation (Stage 2).
ϕ_s	Encoder for sensor s , $\phi_s : \mathbb{R}^{B_s \times P \times P} \rightarrow \mathbb{R}^d$.
$\mathbf{e}_i(t)$	Embedding, $\mathbf{e}_i(t) = \phi_s(X_i(t))$.
$\mathbf{h}(t)$	Harmonic design row, $\in \mathbb{R}^m$, $m = 2 + 2K$.
Θ_i	Harmonic coefficient matrix for tile i , $\in \mathbb{R}^{m \times d}$.
$\mathbf{r}_i(t)$	First-stage residual (tile effect removed).
$\tilde{\mathbf{r}}_i(t)$	Second-stage residual (time effect also removed).
$s_i(t)$	Nonconformity score.
$p_i(t)$	Conformal p-value.
$\mathcal{C}(i)$	Control cohort for tile i .
$\kappa(i, t)$	Mondrian stratum of observation (i, t) .
α	Target per-decision false-alarm rate.
K	Number of harmonic orders.
ω	Fundamental angular frequency, $2\pi/365.25 \text{ d}^{-1}$.

4.2 Task definitions

Definition 3 (Semantic retrieval, R1.1–R1.4). Given a query q (free text or exemplar image) and a metadata predicate Π (AOI, date range, sensor), return a ranked list π over $\{(i, t)\}$ satisfying Π , ordered by a relevance score $\sigma(q, i, t)$.

Definition 4 (Change detection, R2.1). Given a tile i and a window $[t_a, t_b]$, decide $\mathcal{H}_1$ (a meaningful change occurred) versus $\mathcal{H}_0$ (no meaningful change occurred; any observed difference is attributable to season, atmosphere, geometry, sensor or registration), and report a calibrated confidence.

Definition 5 (Change typing, R2.2). Given $\mathcal{H}_1$, return a prediction set $\hat{Y}_i \subseteq \{\text{construction, clearance, water-extent, road development, other}\}$ with a coverage guarantee $\mathbb{P}(Y_i \in \hat{Y}_i) \geq 1 - \alpha_{\text{cls}}$.

Definition 6 (Observability-bounded dating, R2.3). Given $\mathcal{H}_1$, return (t^-, t^+, G) where t^- is the last usable observation consistent with $\mathcal{H}_0$, t^+ is the first usable observation at which $\mathcal{H}_1$ is confirmed, and $G = \{t \in (t^-, t^+) : q_i(t) < q_{\min}\}$ enumerates the acquisitions that existed but were not usable.

Remark 7. Definition 4 is the formalisation of “the earliest available observation at which the change is supported by usable imagery”. Note that t^+ is the answer to the question as asked; G is the honest statement of why the answer is not earlier; and $[t^-, t^+]$ is the true uncertainty. Systems that report a single date are answering a different, easier question.

4.3 The nuisance generative model

We adopt the following structural model, which is the formal backbone of the entire design.

$$\mathbf{e}_i(t) = \boldsymbol{\alpha}_i + \boldsymbol{\beta}_i t + f_i(t) + \boldsymbol{\tau}_\kappa(t) + \mathbf{c}_i(t) + \boldsymbol{\eta}_i(t) \quad (1)$$

- $\boldsymbol{\alpha}_i \in \mathbb{R}^d$: the *location effect*. What this place fundamentally is. Time-invariant.
- $\boldsymbol{\beta}_i t$: slow secular *trend* — gradual urban densification, forest growth, reservoir siltation.
- $f_i(t)$: the *phenological cycle*, periodic with period one year: $f_i(t + 365.25) = f_i(t)$. Crop cycles, monsoon greening, seasonal water extent, snow cover. *This is the dominant nuisance term and the reason bi-temporal methods fail.*
- $\boldsymbol{\tau}_\kappa(t)$: the *acquisition effect*, shared by all tiles in stratum κ at time t . Residual atmospheric haze after correction, solar illumination geometry, view angle, sensor calibration state, processing baseline changes. *This is the term that pseudo-invariant-feature normalisation attacks, and it is exactly a time fixed effect.*
- $\mathbf{c}_i(t)$: the *signal*. Real, persistent, physical change on the ground.
- $\boldsymbol{\eta}_i(t)$: residual noise — sensor noise, speckle, sub-pixel misregistration, encoder stochasticity.

The entire design follows from Eq. 1. $\boldsymbol{\alpha}_i + \boldsymbol{\beta}_i t + f_i(t)$ is a *tile fixed effect*, estimable from that tile’s own history. $\boldsymbol{\tau}_\kappa(t)$ is a *time fixed effect*, estimable from contemporaneous observations of comparable tiles. Removing both identifies $\mathbf{c}_i(t)$ up to noise. This is a two-way fixed-effects model, and change detection is a difference-in-differences problem. To our knowledge this framing is new to the remote-sensing change-detection literature, and it is not a rhetorical device: it is the correct estimator for this confounding structure, and it tells us exactly which assumptions we need and how to test them.

5 Theory

5.1 Harmonic Embedding Fields

5.1.1 Model

Let K be the harmonic order and define the design row

$$\mathbf{h}(t) = [1, \bar{t}(t), \cos \omega t, \sin \omega t, \cos 2\omega t, \sin 2\omega t, \dots, \cos K\omega t, \sin K\omega t] \in \mathbb{R}^m, \quad m = 2 + 2K, \quad (2)$$

where $\omega = 2\pi/365.25 \text{ d}^{-1}$ and $\bar{t}(t) = (t - t_0)/\Delta T$ is the time index centred and scaled to $[-1, 1]$ for numerical conditioning. Then

$$\mathbf{e}_i(t) = \mathbf{h}(t) \boldsymbol{\Theta}_i + \varepsilon_i(t), \quad \boldsymbol{\Theta}_i \in \mathbb{R}^{m \times d}. \quad (3)$$

The rows of $\boldsymbol{\Theta}_i$ carry direct interpretation: row 1 is the location effect α_i , row 2 the trend β_i , and rows $3 \dots m$ the phenological cycle f_i in cosine/sine basis.

5.1.2 Robust weighted ridge estimation

Stack $n_i = |\mathcal{T}_i|$ observations into $\mathbf{H}_i \in \mathbb{R}^{n_i \times m}$ and $\mathbf{E}_i \in \mathbb{R}^{n_i \times d}$, with weight matrix $\mathbf{W}_i = \text{diag}(q_i(t_1), \dots, q_i(t_{n_i}))$ from the quality layer. Solve

$$\hat{\boldsymbol{\Theta}}_i = \arg \min_{\boldsymbol{\Theta}} \|\mathbf{W}_i^{1/2}(\mathbf{E}_i - \mathbf{H}_i \boldsymbol{\Theta})\|_F^2 + \lambda \|\boldsymbol{\Lambda} \boldsymbol{\Theta}\|_F^2, \quad (4)$$

with the closed-form solution

$$\hat{\boldsymbol{\Theta}}_i = (\mathbf{H}_i^\top \mathbf{W}_i \mathbf{H}_i + \lambda \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda})^{-1} \mathbf{H}_i^\top \mathbf{W}_i \mathbf{E}_i. \quad (5)$$

The penalty matrix. $\boldsymbol{\Lambda} = \text{diag}(0, 0, 1, 1, 2^\nu, 2^\nu, \dots, K^\nu, K^\nu)$ with $\nu \approx 2$ penalises high-order harmonics quadratically. This is the mechanism that prevents the well-documented CCDC failure mode in which high-order terms overfit gaps in the observation record and generate spurious breaks. It plays the role of the Harmonic Adaptive Penalty Operator, in closed form.

Robustification. Cloud, haze and shadow that survive masking are outliers with respect to Eq. 1, and least squares has breakdown point 0. We therefore run iteratively reweighted least squares with the Huber ψ -function: at iteration ℓ , update

$$w_i^{(\ell+1)}(t) = q_i(t) \cdot \psi_\delta \left(\frac{\|\mathbf{e}_i(t) - \mathbf{h}(t) \hat{\boldsymbol{\Theta}}_i^{(\ell)}\|}{\hat{\varsigma}^{(\ell)}} \right), \quad \psi_\delta(u) = \min \left(1, \frac{\delta}{|u|} \right), \quad (6)$$

with $\hat{\varsigma}$ a robust scale (median absolute deviation) and $\delta = 1.345$ for 95% Gaussian efficiency. Three to five iterations suffice in practice.

Proposition 8 (Identifiability). *$\hat{\boldsymbol{\Theta}}_i$ is uniquely determined if $\text{rank}(\mathbf{W}_i^{1/2} \mathbf{H}_i) = m$, which requires at least $m = 2 + 2K$ observations with $q_i(t) > 0$ at distinct phases. With $\lambda > 0$ the ridge term guarantees a unique solution even under rank deficiency, at the price of shrinkage toward zero seasonality.*

Practical order selection. We use $K = 2$ (annual and semi-annual, $m = 6$) as the default. Semi-annual capture is important in India, where the double-cropping cycle (kharif and rabi) produces a genuinely bimodal annual NDVI signature that a single harmonic cannot represent. We require $n_i \geq 3m = 18$ usable observations spanning ≥ 1.5 years for a full fit; below that we fall back per §7.1. K is selected per stratum by generalised cross-validation on held-out observations.

Remark 9 (Why harmonic least squares and not BFAST). BFAST requires regularly spaced observations. After cloud masking, our series is emphatically irregular — that irregularity is the whole point of R2.3. Harmonic regression on explicit timestamps handles arbitrary sampling natively, admits per-observation quality weights, and has a closed-form solution that admits recursive update. These three properties are exactly what R6.2 (incremental ingestion without rebuild) demands.

5.1.3 Derived representations

From $\hat{\Theta}_i$ we read off three products used throughout the system.

(a) Season-invariant semantic vector. Because $\int_0^{2\pi/\omega} \cos(k\omega t) dt = \int_0^{2\pi/\omega} \sin(k\omega t) dt = 0$ for all $k \geq 1$, the mean of the fitted trajectory over a full cycle is simply

$$\mathbf{v}_i^{\text{sem}} = \hat{\alpha}_i + \hat{\beta}_i \bar{t}_{\text{mid}}. \quad (7)$$

This is a representation of the tile with the seasonal cycle *analytically removed*, not merely averaged over. It is what we index for retrieval, and it eliminates a real and rarely acknowledged retrieval pathology: with conventional single-date indexing, the answer to the query “river” varies with which date happened to be indexed.

(b) Phenological fingerprint. Convert the cosine/sine pairs to amplitude–phase form, which is more interpretable and more stable under small timing perturbations:

$$A_{ik} = \|\hat{\gamma}_{ik}\| \oplus \|\hat{\delta}_{ik}\|, \quad \varphi_{ik} = \text{atan2}(\hat{\delta}_{ik}, \hat{\gamma}_{ik}), \quad \mathbf{v}_i^{\text{phen}} = [A_{i1}, \varphi_{i1}, \dots, A_{iK}, \varphi_{iK}]. \quad (8)$$

A is the strength of seasonality (irrigated double-cropped land has high A_1 and A_2 ; a paved industrial site has near-zero A); φ is its timing (which encodes crop calendar, monsoon onset, snowmelt date). **This vector is a discriminative site descriptor that no single-timestamp embedding can express**, and it is what makes R4.1/R4.2 (discovery and clustering) work well rather than merely work.

(c) Residual. $\mathbf{r}_i(t) = \mathbf{e}_i(t) - \mathbf{h}(t)\hat{\Theta}_i$, the first-stage difference in §5.3.

5.2 Operating modes: the guarantee does not require a long time series

A reader may reasonably object that §5.1 presupposes a multi-year archive, and that the organisers’ evaluation window may be short — or that the held-out change set may consist of bi-temporal *pairs*, in which case a harmonic model cannot be fitted at all. We take that objection seriously enough to design for it rather than relegate it to a fallback.

The load-bearing observation. C2 (the difference-in-differences estimator, §5.3) and C3–C4 (conformal calibration and sequential control, §§5.4–5.5) *do not depend on C1*. They operate on a residual and a score, and are indifferent to how the residual was produced. The harmonic field is the best available residual generator when a series exists; when it does not, the statistical apparatus — which is the principal contribution — transfers intact to a plain bi-temporal difference.

The system therefore runs in three modes, selected **per tile** from the effective sample size n_i^{eff} and temporal span, so that a single run may operate in different modes in different parts of the same AOI.

Mode	Precondition	Behaviour
A — Full	$\text{span} \geq 1.5 \text{ yr},$ $n^{\text{eff}} \geq 3m$	Harmonic field at $K = 2$. All of C1–C4. Phenological fingerprint available, so cohort matching and discovery run on estimated seasonal response.
B — Reduced	$\text{span } 0.5\text{--}1.5 \text{ yr}$	$K = 1$ (annual term only), heavier ridge λ , and where available a seasonal prior transferred from tiles in the same $\mathbf{v}^{\text{phen}}$ cluster with longer records. Fingerprint is low-confidence and is down-weighted in cohort matching.
C — Bi-temporal	two epochs only	No harmonic model. Residual is the harmonised bi-temporal difference $\mathbf{r}_i = \mathbf{e}_i(t_2) - \mathbf{e}_i(t_1)$. C2 and C3 operate unchanged.

Table 4: Operating modes. Mode is selected per tile, not per run.

Mode C in detail. Three substitutions make the bi-temporal path fully operational:

1. **Cohort matching covariate.** With no fingerprint available, the control cohort $\mathcal{C}(i)$ is matched on the *pre-epoch* semantic vector $\mathbf{e}_i(t_1)$ together with elevation and aspect. This is a weaker covariate than $\mathbf{v}^{\text{phen}}$ — it matches on what a place looks like rather than on how it responds to seasonal forcing — but it remains a legitimate match on observable confounders.
2. **The DiD does more work.** In Mode A the harmonic fit removes phenology and the cohort removes the acquisition effect. In Mode C the cohort must absorb *both*, since the seasonal difference between t_1 and t_2 is common to all tiles in the stratum and therefore enters τ_κ . This is precisely the estimand that Hamunyela et al. [3] showed works, and it is why their spatially normalised detector outperformed an explicit seasonal model in strongly seasonal dry forest.
3. **Stratification absorbs the epoch pair.** The Mondrian stratum gains the season-pair as a component, $\kappa = (\ell_i, \text{season}(t_1) \times \text{season}(t_2), \text{sensor-pair}, \varrho_i)$, so an April–August comparison is calibrated only against other April–August comparisons.

What is lost, stated plainly. Mode C forfeits the phenological fingerprint (weakening discovery, §7), forfeits sub-annual change dating (§5.6 degenerates to “between t_1 and t_2 ”, though the observability gap G remains computable from the acquisition record whenever the archive is present), and forfeits the anytime-valid e-process, which is unnecessary with a single look — plain split conformal suffices and Theorem 12 applies directly. What is *retained* is the calibrated false-alarm rate, the confounder-conditional validity, and the queue-level FDR bound. Those are the claims we intend to defend, and they survive the worst case.

5.3 Change as a difference-in-differences estimand

5.3.1 Two-stage differencing

Substituting Eq. 1 into the residual definition, and noting that the harmonic fit estimates $\alpha_i + \beta_i t + f_i(t)$:

$$\mathbf{r}_i(t) = \mathbf{e}_i(t) - \mathbf{h}(t)\hat{\Theta}_i \approx \tau_\kappa(t) + \mathbf{c}_i(t) + \boldsymbol{\eta}_i(t). \quad (9)$$

The tile effect is gone. The acquisition effect $\tau_\kappa(t)$ remains, and it is not small: a hazy acquisition, an off-nadir pass, a processing-baseline change, or an anomalously early monsoon displaces *every* tile in the scene simultaneously. Left untreated it produces correlated false alarms across an entire acquisition — the worst possible failure mode for an analyst queue, because it floods the queue in a single event.

We estimate it from a control cohort $\mathcal{C}(i)$ and difference again:

$$\tilde{\mathbf{r}}_i(t) = \mathbf{r}_i(t) - \text{med}_{m \in \mathcal{C}(i)} \mathbf{r}_m(t) \quad (10)$$

5.3.2 Control cohort construction

$\mathcal{C}(i)$ must satisfy: same acquisition (hence same τ), comparable land surface (hence comparable response to τ), and predominantly unchanged. We construct it as

$$\mathcal{C}(i) = \arg \text{top-}^k_{m \neq i} \left\{ -\|\mathbf{v}_m^{\text{phen}} - \mathbf{v}_i^{\text{phen}}\| \mid m \in \text{same acquisition, } |z_m - z_i| < \epsilon_z, |\theta_m - \theta_i| < \epsilon_\theta \right\} \quad (11)$$

i.e. the k nearest neighbours of tile i in *phenological-fingerprint space*, restricted to the same acquisition and to a similar elevation z and terrain aspect θ band (both from CartoDEM). Typical $k = 64$.

Why matching on phenology is the right matching. In causal-inference terms we need to match on the confounder. Here the confounder is how a surface responds to seasonal and atmospheric forcing — and $\mathbf{v}^{\text{phen}}$ is an estimate of precisely that response function, learned from data. We are therefore matching on an estimated response-to-*nuisance*, which is the correct covariate. Contrast this with Hamunyela et al. [3], whose spatial-context normalisation uses *geographic adjacency* as a proxy. Adjacency is a good proxy in homogeneous forest and a poor one at a land-cover boundary, and their own stated limitation — that cloud masking removes reference pixels and destabilises the cohort — follows directly from tying the cohort to fixed geography. Our cohort is defined in feature space, so a masked member is simply replaced by the next nearest neighbour and the estimator degrades gracefully.

5.3.3 Identification assumptions

The estimator $\tilde{\mathbf{r}}_i(t)$ recovers $\mathbf{c}_i(t)$ under three assumptions, which we state explicitly because stating them is what separates an estimator from a heuristic.

Assumption 1 (Common shock / parallel trends). $\tau_m(t) = \tau_\kappa(t)$ for all $m \in \mathcal{C}(i) \cup \{i\}$. Justification: atmospheric state, solar geometry and sensor state vary smoothly over a scene at scales far exceeding a 2.24 km tile, and cohort members are drawn from the same acquisition and similar terrain. *Testable*: regress $\mathbf{r}_m(t)$ on distance-to- i within the cohort; a significant gradient indicates violation and triggers cohort narrowing.

Assumption 2 (Control purity). Fewer than half of $\mathcal{C}(i)$ experience real change at t . Justification: change is rare — in DynamicEarthNet only 1.28% of pixels change month-to-month, and in MUDS 0.03% [26]. The median in Eq. 10 has breakdown point 1/2, so the estimator tolerates contamination up to that level. *Testable*: report the cohort inlier fraction as a per-decision diagnostic; a low fraction flags a region-wide event (flood, fire) that should be escalated rather than differenced away.

Assumption 3 (No anticipation). $\mathbf{c}_i(t) = 0$ for $t < t^*$, the true change time. Justification: physical change on the ground does not appear in imagery before it happens. Site preparation preceding construction is a genuine earlier change, not an anticipation effect, and we *want* to detect it.

Remark 10 (The region-wide event case). Assumption 2 has a deliberate and important consequence. If a flood inundates an entire district, the cohort floods too, and the two-way estimator will correctly report “nothing anomalous *relative to comparable places*”. This is not a bug — it is the estimand doing its job — but it is not what the analyst wants. We therefore retain and surface

the first-stage residual $\mathbf{r}_i(t)$ alongside $\tilde{\mathbf{r}}_i(t)$. Large $\|\mathbf{r}\|$ with small $\|\tilde{\mathbf{r}}\|$ is the signature of a *regional* event and is routed to a separate queue. This decomposition, which comes for free from the two-stage structure, is itself an operationally useful distinction that single-stage methods cannot make.

5.4 Conformal change control

5.4.1 Nonconformity score

Estimate the residual covariance $\hat{\Sigma}_\kappa$ per stratum from stable segments (shrunk toward a diagonal target by Ledoit–Wolf for stability when d is large relative to sample count) and define

$$s_i(t) = \sqrt{\tilde{\mathbf{r}}_i(t)^\top \hat{\Sigma}_\kappa^{-1} \tilde{\mathbf{r}}_i(t)}. \quad (12)$$

Mahalanobis rather than Euclidean, because embedding dimensions are neither isotropic nor independent; using Euclidean distance silently over-weights the highest-variance directions, which are typically the most season-sensitive.

5.4.2 Label-free null calibration

Split conformal requires a calibration set drawn from $\mathcal{H}_0$. Obtaining labelled no-change examples is normally expensive. Our decomposition supplies them for free:

1. Fit the harmonic model on the full series.
2. Identify the longest *stable segment* per tile: the maximal contiguous run of usable observations containing no candidate break under a loose screening threshold.
3. Every observation in a stable segment is, by construction, a sample from $\mathcal{H}_0$. Harvest its score s into $\mathcal{D}_{\text{cal}}$.
4. Additionally harvest cross-year same-phase pairs: (i, t) and $(i, t + 365.25)$ both in stable segments give a matched null under identical phenological phase.

This yields $|\mathcal{D}_{\text{cal}}|$ on the order of $N \times T \times (1 - \text{change rate})$ — in our worked example, millions of null samples, entirely without annotation.

5.4.3 The circularity, and why it is a bounded trade rather than a hole

Step 2 defines the calibration set using a screening threshold derived from the very detector being calibrated. This circularity is real and must be confronted: if the screen misses changes, the “null” set is contaminated with genuine changes, and Theorem 12 calibrates against a distribution that is not $\mathcal{H}_0$.

The following shows the contamination pushes in the *safe* direction.

Lemma 11 (Contamination is conservative). *Let calibration scores be drawn from the mixture $F_{\text{mix}} = (1 - \varepsilon)F_0 + \varepsilon F_1$, where F_0 is the null score distribution, F_1 that of undetected changes, and ε the contamination rate. Assume changes are stochastically larger, $F_1(x) \leq F_0(x)$ for all x . Let $\hat{\tau}_\alpha = F_{\text{mix}}^{-1}(1 - \alpha)$ be the realised threshold and $\alpha_{\text{eff}} = 1 - F_0(\hat{\tau}_\alpha)$ the true false-alarm rate it delivers. Then*

$$\frac{\alpha - \varepsilon}{1 - \varepsilon} \leq \alpha_{\text{eff}} \leq \alpha.$$

Proof. By definition $(1 - \varepsilon)F_0(\hat{\tau}_\alpha) + \varepsilon F_1(\hat{\tau}_\alpha) = 1 - \alpha$, so $F_0(\hat{\tau}_\alpha) = [1 - \alpha - \varepsilon F_1(\hat{\tau}_\alpha)] / (1 - \varepsilon)$ and hence

$$\alpha_{\text{eff}} = 1 - F_0(\hat{\tau}_\alpha) = \frac{\alpha - \varepsilon(1 - F_1(\hat{\tau}_\alpha))}{1 - \varepsilon}.$$

Stochastic ordering gives $1 - F_1(\hat{\tau}_\alpha) \geq 1 - F_0(\hat{\tau}_\alpha) = \alpha_{\text{eff}}$, so $\alpha_{\text{eff}}(1 - \varepsilon) \leq \alpha - \varepsilon\alpha_{\text{eff}}$, i.e. $\alpha_{\text{eff}} \leq \alpha$. The lower bound follows from $1 - F_1(\hat{\tau}_\alpha) \leq 1$. $\square$

Reading of Lemma 11. Contaminating a calibration set with undetected changes *raises* the threshold, because the contaminants sit in the upper tail we are taking the quantile of. Validity is therefore preserved exactly — $\alpha_{\text{eff}} \leq \alpha$ — and the entire cost is paid in *power*. The bound is tight enough to be actionable: change is rare, at 1.28% month-to-month in DynamicEarthNet and 0.03% in MUDS [26], so with $\varepsilon = 0.013$ and $\alpha = 0.05$ the effective level lies in $[0.037, 0.050]$ — a modest loss of sensitivity in exchange for a guarantee that still holds. **The essential condition is $\varepsilon < \alpha$** ; if contamination approaches the target level the detector silently stops firing, which is why we monitor ε rather than assume it.

Two practical consequences. First, we estimate ε per stratum by holding out a slice of *analyst-confirmed rejections* (Stage 10) as an uncontaminated audit set, and report the implied α_{eff} band alongside the nominal α . Second, the screening threshold in step 2 should be set *permissively*: excluding a marginal observation from $\mathcal{D}_{\text{cal}}$ costs a little calibration sample, whereas admitting a genuine change costs power. Erring toward exclusion is the correct asymmetry, and it is the opposite of what one would choose if the screen were a detector rather than a filter.

5.4.4 Validity

Theorem 12 (Marginal conformal validity). *Let $\{s_1, \dots, s_n\}$ be nonconformity scores on $\mathcal{D}_{\text{cal}}$ and let s_{n+1} be the score of a new observation. If $s_1, \dots, s_{n+1}$ are exchangeable under $\mathcal{H}_0$, then setting*

$$\hat{\tau}_\alpha = s_{(\lceil (n+1)(1-\alpha) \rceil)} \quad (\text{the } \lceil (n+1)(1-\alpha) \rceil\text{-th order statistic})$$

gives $\mathbb{P}(s_{n+1} > \hat{\tau}_\alpha \mid \mathcal{H}_0) \leq \alpha$. Equivalently, the conformal p-value

$$p_{n+1} = \frac{1 + |\{j \leq n : s_j \geq s_{n+1}\}|}{n+1}$$

satisfies $\mathbb{P}(p_{n+1} \leq \alpha \mid \mathcal{H}_0) \leq \alpha$ for all $\alpha \in (0, 1)$.

Proof. Under exchangeability the rank R of s_{n+1} among $\{s_1, \dots, s_{n+1}\}$ is uniformly distributed on $\{1, \dots, n+1\}$ (ties broken at random). The event $\{s_{n+1} > \hat{\tau}_\alpha\}$ is contained in $\{R > \lceil (n+1)(1-\alpha) \rceil\}$, which has probability $(n+1 - \lceil (n+1)(1-\alpha) \rceil)/(n+1) \leq \alpha$. The p-value statement follows since $p_{n+1} \leq \alpha \iff R \geq (n+1)(1-\alpha) + 1$. $\square$

The guarantee is *distribution-free* (no assumption on the embedding distribution), *finite-sample* (no asymptotics), and *model-agnostic* (any encoder, any d). This is the statistical property that Requirement R3.3 has been asking for all along.

5.4.5 Mondrian stratification — handling the exchangeability break

Theorem 12 requires exchangeability, and in this domain exchangeability is *false* in the raw: a summer Sentinel-2 observation over irrigated cropland and a winter Landsat observation over bare rock are not exchangeable. This is precisely why the Earth-observation community has flagged conformal change detection as an open problem rather than solved it [30].

The standard and correct remedy is Mondrian (class-conditional) conformal prediction: partition the observation space into strata within which exchangeability is plausible, and calibrate independently in each. We stratify on

$$\kappa(i, t) = (\underbrace{\ell_i}_{\text{land-cover class}}, \underbrace{\lfloor \text{doy}(t)/\Delta \rfloor}_{\text{season bucket}}, \underbrace{s(t)}_{\text{sensor}}, \underbrace{\varrho_i}_{\text{relief class}}) \quad (13)$$

with ℓ_i from an unsupervised clustering of $\mathbf{v}^{\text{phen}}$ (no external labels needed), $\Delta = 30$ days, and ϱ_i a three-level terrain-ruggedness class from CartoDEM.

Corollary 13 (Conditional validity). *Applying Theorem 12 within stratum κ yields $\mathbb{P}(\text{flag} \mid \mathcal{H}_0, \kappa) \leq \alpha$ for every κ with sufficient calibration samples.*

This is a strictly stronger and far more operationally meaningful guarantee than marginal validity. Marginal validity permits a system to achieve 5% overall false alarms while producing 40% false alarms in the monsoon and 0% in the dry season. Conditional validity forbids that. **Given that the held-out evaluation set will contain deliberately hard confounder cases, conditional validity is what will actually be measured.**

For strata with insufficient samples we fall back to a hierarchical pooling scheme, borrowing calibration from the parent stratum and reporting the effective sample size as a confidence qualifier. We additionally implement conditional-validity corrections (Dvoretzky–Kiefer–Wolfowitz and Simes), which are documented to give tighter false-alarm control than marginal conformal when calibration data are scarce — exactly our regime in rare strata.

5.4.6 False discovery rate over the review queue

Per-decision validity is not the same as queue-level quality. If we test $M = 10^6$ tile-dates at $\alpha = 0.05$, we expect 5×10^4 false alarms even under a perfectly valid procedure. The analyst does not care about per-test error; the analyst cares about what fraction of the review queue is junk.

We therefore apply the Benjamini–Yekutieli procedure to the conformal p-values, which controls the false discovery rate under *arbitrary* dependence:

$$\text{reject } H_{(1)}, \dots, H_{(k^*)}, \quad k^* = \max \left\{ k : p_{(k)} \leq \frac{k}{M \cdot c(M)} q \right\}, \quad c(M) = \sum_{j=1}^M \frac{1}{j}. \quad (14)$$

BY rather than Benjamini–Hochberg because our p-values are strongly spatially and temporally correlated (adjacent tiles overlap by 50%; consecutive dates share the same fitted model), and BH requires positive regression dependence which we cannot establish. The $c(M) \approx \ln M + \gamma$ factor costs power, which we accept: R3.3 explicitly prefers precision.

Corollary 14 (Queue-level guarantee). $\mathbb{E}[FDP] \leq q$, where FDP is the proportion of the analyst’s review queue consisting of false alarms.

This converts into a statement an analyst can act on: “at $q = 0.10$, no more than one item in ten of this queue is expected to be spurious.” We know of no other Earth-observation system that can make this statement.

5.5 Anytime-valid sequential detection

5.5.1 The repeated-looking problem

Requirement R6.2 mandates incremental ingestion. That creates a statistical hazard that is easy to miss: every new acquisition is another look at the same hypothesis. A fixed-sample test applied repeatedly has false-alarm probability tending to 1 — with enough acquisitions, *every* tile eventually crosses any fixed threshold. An archive that ingests forever and tests at every ingestion will, under a naive procedure, eventually flag everything.

The correct frame is anytime-valid inference.

5.5.2 Why the naive e-process does not work

The tempting construction is to convert each conformal p-value into an e-value with a calibrator $\varsigma : [0, 1] \rightarrow [0, \infty]$ satisfying $\int_0^1 \varsigma(p) dp \leq 1$, multiply them, and appeal to Ville’s inequality. That argument requires each $p_i(t)$ to be *conditionally* super-uniform given the past. Theorem 12 does not supply this: the rank argument gives *marginal* validity only, and marginal validity does not imply conditional validity.

The distinction is not academic here. Consecutive residuals at a tile are dependent for three concrete reasons: (i) a haze or shadow event spans several acquisitions; (ii) every residual at tile i is computed against *the same* fitted $\hat{\Theta}_i$, so a fitting error correlates all of them; and (iii) the control cohort is stable across nearby dates. Under such dependence the naive product is not a supermartingale and the guarantee is void. We therefore construct the detector so that the required exchangeability actually holds, rather than assuming it.

5.5.3 Construction: a conformal test martingale

Three modifications, each addressing one of the dependence sources above.

(a) Out-of-fold residuals. Partition the observation indices of tile i into J folds and compute the residual at t against a model fitted *without* the fold containing t :

$$\mathbf{r}_i(t) = \mathbf{e}_i(t) - \mathbf{h}(t) \hat{\Theta}_i^{(-J(t))}. \quad (15)$$

No observation contributes to the model against which it is scored, which removes dependence source (ii) entirely. The cost is J fits per tile; since a fit costs microseconds (§5.1), this is free in practice.

(b) Smoothed conformal p-values. With $\tau \sim \text{U}(0, 1)$ drawn independently,

$$p_i(t) = \frac{|\{j : s_j > s_i(t)\}| + \tau |\{j : s_j = s_i(t)\}| + \tau}{n + 1}, \quad j \in \mathcal{D}_{\text{cal}}^\kappa. \quad (16)$$

Smoothing makes the p-values *exactly* uniform rather than conservative, which is what upgrades the supermartingale to a martingale and recovers the power lost to ties.

(c) Predictable betting functions. Replace the fixed calibrator with a sequence $\varsigma_t : [0, 1] \rightarrow [0, \infty)$ with $\int_0^1 \varsigma_t(u) du \leq 1$, each ς_t measurable with respect to $\mathcal{F}_{t-1}$. We use the power family $\varsigma_\varkappa(p) = \varkappa p^{\varkappa-1}$, $\varkappa \in (0, 1)$, with $\varkappa$ re-estimated online from past p-values — legitimate precisely because predictability is all the martingale property requires. Then

$$E_i(t) = \prod_{\substack{t' \leq t \\ q_i(t') \geq q_{\min}}} \varsigma_{t'}(p_i(t')), \quad E_i(t_0) = 1, \quad (17)$$

and a change is confirmed the first time $E_i(t) \geq 1/\alpha$.

Assumption 4 (Null exchangeability). Under $\mathcal{H}_0$ at tile i , the sequence of out-of-fold nonconformity scores $(s_i(t))_{t: q_i(t) \geq q_{\min}}$ is exchangeable with the calibration scores of its Mondrian stratum κ .

Theorem 15 (Anytime validity). *Under Assumption 4, $(E_i(t))_t$ is a nonnegative martingale with $E_i(t_0) = 1$, and by Ville's inequality*

$$\mathbb{P}(\exists t : E_i(t) \geq 1/\alpha \mid \mathcal{H}_0) \leq \alpha.$$

Proof. Under exchangeability, smoothed conformal p-values produced by the online (transductive) scheme are i.i.d. $\text{U}(0, 1)$ — this is the exchangeability-martingale result of Vovk et al. [28, Ch. 7]. Since ς_t is $\mathcal{F}_{t-1}$ -measurable and $p_i(t) \perp \mathcal{F}_{t-1}$ with $p_i(t) \sim \text{U}(0, 1)$,

$$\mathbb{E}[\varsigma_t(p_i(t)) \mid \mathcal{F}_{t-1}] = \int_0^1 \varsigma_t(u) du \leq 1,$$

so $\mathbb{E}[E_i(t) \mid \mathcal{F}_{t-1}] \leq E_i(t-1)$, with equality when $\int \varsigma_t = 1$. Hence $(E_i(t))$ is a nonnegative martingale started at 1, and Ville's inequality for nonnegative supermartingales gives $\mathbb{P}(\sup_t E_i(t) \geq 1/\alpha) \leq \alpha$. $\square$

Remark 16 (Assumption 4 is testable, and we test it). Modifications (a) and (b) remove the dependence induced by the shared fit and by ties. What they do not remove is genuine physical autocorrelation — source (i). We therefore treat Assumption 4 as an empirical claim: we run a Ljung–Box test on the out-of-fold residual series per stratum and report the result alongside the coverage curve. Where autocorrelation survives at lag ℓ , we thin the e-process to every ℓ -th usable acquisition, which restores exchangeability at the cost of detection latency. Reporting the test rather than asserting the assumption is the difference between a guarantee and a claim.

Why this matters more than it may appear. Theorem 15 says the false-alarm guarantee holds *at every time step simultaneously*, for an *unbounded* number of looks, with *no* multiple-testing correction over time and *no* recalibration on ingestion. The system may ingest acquisitions forever and the guarantee never degrades. This is exactly the property an operational, incrementally ingesting archive requires, and it is the natural statistical counterpart to Requirement R6.2. It also replaces the arbitrary “ m consecutive exceedances” persistence rule used by CCDC and its descendants with a principled evidence accumulator: transient artefacts contribute e-values near 1 and decay from the product; genuine persistent change compounds multiplicatively and crosses the boundary.

5.6 Observability-bounded change dating

Let $t^\dagger = \min\{t : E_i(t) \geq 1/\alpha\}$ be the confirmation time. Define:

$$t^- = \max\{t < t^\dagger : q_i(t) \geq q_{\min} \wedge p_i(t) > \alpha_{\text{loose}}\} \quad \text{last clean null observation} \quad (18)$$

$$t^+ = \min\{t \leq t^\dagger : q_i(t) \geq q_{\min} \wedge p_i(t) \leq \alpha_{\text{loose}}\} \quad \text{first usable supporting observation} \quad (19)$$

$$G = \{t \in (t^-, t^+) : \exists \text{ acquisition at } t \wedge q_i(t) < q_{\min}\} \quad \text{observability gap} \quad (20)$$

The system reports the triple (t^-, t^+, G) together with a changepoint posterior $\mathbb{P}(t^* = t)$ obtained by Bayesian model averaging over segmentations of the residual series (BEAST-style [2]), restricted to $[t^-, t^+]$.

Example output.

Change confirmed. Last clean pre-change observation: 2025-03-12 (S2A, 4% cloud). First usable supporting observation: 2025-06-26 (S2B, 11% cloud). Posterior mass over onset concentrated 2025-04-18 to 2025-06-26. Observability gap: 7 acquisitions between those dates existed but were unusable (5 cloud > 70%, 2 sensor gap). Corroborating S1 backscatter break at 2025-05-04 narrows the interval to 2025-04-18 – 2025-05-04.

That last sentence illustrates a valuable emergent property: because SAR is cloud-immune, it can narrow an optical observability gap. The SAR residual series is fused into the same e-process, so this happens automatically rather than as a special case.

5.7 Change morphology: appearance, disappearance, expansion, contraction

Requirement R2.1 names four change kinds. Appearance and disappearance are readable from the sign of the residual, but *expansion* and *contraction* are statements about the changing **extent of an object**, and a tile-level detector has no objects. This section supplies the missing object layer. It is deliberately built from components the system already has, so it adds one module rather than a second pipeline.

5.7.1 From tiles to change objects

1. **Patch-resolution residual field.** At a confirmed break, evaluate the residual at patch granularity (16×16 px tokens, i.e. 160 m) rather than pooling to the tile vector. This reuses exactly the patch embeddings computed for late-interaction reranking (§7), so it costs nothing new.
2. **Mosaic across the overlap.** With 50% tile overlap every patch is covered by up to four tiles. Combine by taking the *median* conformal p-value across covering tiles. Median rather than minimum: the minimum would multiply the effective number of tests per patch and quietly inflate the false-alarm rate, whereas the median is robust and does not.
3. **Threshold and connect.** Threshold the mosaicked p-field at the stratum’s conformal level and extract connected components (8-connectivity), yielding change *objects* with polygon geometry Ω_c , area A_c , and an aggregated change signature $\bar{\Delta}_c$.
4. **Vectorise and simplify.** Polygonise in the tile CRS, apply Douglas–Peucker at half the patch size, and record area in projected units. The object polygon — not the tile — becomes the exported geometry, which materially improves the granularity of what reaches the analyst and hedges the risk that ground truth is polygonal rather than tile-level.

5.7.2 Tracking objects across epochs

Let $\mathcal{O}(t)$ be the object set at confirmed epoch t . Match $\mathcal{O}(t_k)$ to $\mathcal{O}(t_{k+1})$ by greedy maximum-IoU assignment, accepting a match when $\text{IoU}(\Omega_c, \Omega_{c'}) \geq \theta_{\text{IoU}}$ (default 0.3; objects grow, so the threshold must be permissive). The classification then reads directly off the match structure and the area trajectory:

Class	Match structure	Additional condition
Appearance	no predecessor	residual sign consistent with acquisition of the new state
Disappearance	no successor; or post-state returns to pre-break signature	$\langle \bar{\Delta}_c, \bar{\Delta}_{c,\text{prev}} \rangle < 0$
Expansion	matched, A increasing	new area contiguous with old: $\Omega_c \supset \Omega_{c'}$ up to erosion
Contraction	matched, A decreasing	$\Omega_c \subset \Omega_{c'}$ up to dilation
Fragmentation / merge	one-to-many / many-to-one	reported as its own class rather than forced into the four

The containment conditions are what separate genuine expansion from an unrelated new object that happens to be adjacent — a distinction pure area comparison cannot make.

5.7.3 Calibrating the area change

Boundary noise makes every matched object’s area fluctuate, so a bare threshold on ΔA would report expansion everywhere. We therefore treat the area change as another test statistic and pass it through the same machinery rather than inventing a second, uncalibrated rule:

$$s_c^A = \frac{A_c(t_{k+1}) - A_c(t_k)}{\frac{1}{2}(A_c(t_{k+1}) + A_c(t_k))}, \quad (21)$$

a symmetric relative area change. Calibrate its null distribution on *stable* objects — those matched across consecutive epochs at tiles with no confirmed break — within the same stratum and size decile, since small objects have proportionally noisier boundaries. Expansion is declared when the conformal p-value of s_c^A falls below α_A , and contraction when the p-value of $-s_c^A$ does.

This closes R2.1 properly: all four named change kinds are decided by calibrated tests rather than by hand-set thresholds, and the size-decile stratification handles the fact that a 10% area change means something very different for a 0.5 km² object than for a 50 km² one. It also produces polygon geometry, which the tile-level design otherwise lacked.

5.8 Change typing with calibrated prediction sets

Given a confirmed break at $t^\dagger$, form the change vector as the difference of stable-segment means:

$$\Delta_i = \frac{1}{|\mathcal{P}^{\text{post}}|} \sum_{t \in \mathcal{P}^{\text{post}}} \mathbf{e}_i(t) - \frac{1}{|\mathcal{P}^{\text{pre}}|} \sum_{t \in \mathcal{P}^{\text{pre}}} \mathbf{e}_i(t). \quad (22)$$

We score change types along three complementary axis families.

(i) **Physical index axes.** Directly interpretable, sensor-grounded, and label-free:

Type	ΔNDVI	ΔNDBI	ΔNDWI	ΔBSI	$\Delta\sigma_{VV}^0$	$\Delta\sigma_{VH}^0$	Reversibility
Construction	↓	↑↑	—	↑	↑↑	↑	permanent
Clearance	↓↓	—	—	↑↑	—	↓↓	permanent
Water-extent gain	↓	↓	↑↑	↓	↓↓	↓↓	often cyclic
Water-extent loss	—	↑	↓↓	↑	↑	↑	often cyclic
Road development	↓ (linear)	↑ (linear)	—	↑	↑	—	permanent

Table 5: Physical signatures used as interpretable axes for change typing.

The *reversibility* column is where our architecture pays a dividend that rule-based systems cannot access. Seasonal water-extent variation is, by definition, captured by $f_i(t)$ and therefore never produces a residual at all. A water-extent change that *does* produce a residual is by construction anomalous relative to that location’s own established hydrological cycle — which is exactly the distinction between “the reservoir filled as it does every July” and “the reservoir filled when it should not have”. Requirement R2.2 asks for water-extent variation as a change class; we deliver it already disambiguated from its own seasonal baseline.

(ii) **Text-anchored semantic axes.** For a vision–language encoder ψ sharing the image space, define zero-shot direction vectors

$$\begin{aligned} \mathbf{u}_{\text{const}} &= \psi(\text{“a construction site with bare soil and buildings”}) \\ &\quad - \psi(\text{“undeveloped vegetated land”}), \end{aligned} \quad (23)$$

and score $\langle \Delta_i, \mathbf{u}_{\text{const}} \rangle / \|\Delta_i\|$. This gives label-free semantic typing and extends to new change types by writing a sentence, which matters because the organisers say “such as” — the class list is open.

(iii) **Geometric axes.** Road development is distinguished by *shape*, not spectrum. We compute the eccentricity and Hough-line response of the connected component of the residual magnitude map and use elongation as a feature.

Calibrated set-valued output. Rather than an arg-max label, we emit a conformal prediction *set*: calibrate per-class nonconformity on the (small) set of analyst-confirmed examples and return $\hat{Y}_i = \{y : p_i^{(y)} > \alpha_{\text{cls}}\}$, with $\mathbb{P}(Y_i \in \hat{Y}_i) \geq 1 - \alpha_{\text{cls}}$. An analyst is better served by “{construction, clearance}, 90% confidence” than by a confident single label that is wrong 30% of the time, and set size is itself a useful triage signal.

5.9 Retrieval theory

5.9.1 The relational-query impossibility

The organisers’ worked query is “newly built structures near a river”. Decompose it:

Fragment	Predicate type	Correct operator
“structures”	semantic, monadic	embedding nearest-neighbour
“newly built”	temporal, monadic	change engine, appearance type
“near a river”	spatial, dyadic (relational)	GIS spatial join

CLIP-family encoders are documented to fail on relational predicates. On the ARO benchmark, CLIP performs at or below chance on Visual Genome Relation [18]; on Winoground it is barely above chance when word order is the only difference [19]; on What’s Up it collapses on spatial relations [20]. The diagnosed causes are structural — the single-vector contrastive bottleneck, the text encoder itself, and the decay of spatial information in deeper layers — not a data-scale problem that domain fine-tuning fixes. Every remote-sensing CLIP variant (RemoteCLIP, GeoRSCLIP, SkyCLIP) inherits the pathology wholesale.

We do not attempt to make the neural model do what it provably cannot. “Near a river” is not a texture; it is a topological relation between a candidate location and a hydrographic feature, and it has an exact, cheap, deterministic answer: **ST_DWithin**. Routing it to PostGIS is not a workaround — it is the correct operator, and it is exact where the neural alternative is at chance.

5.9.2 Query algebra

We define a typed algebra over scored tile-date sets. Let $\mathcal{U} = \mathcal{G} \times \mathcal{T}$ and let a *scored set* be a map $\mathcal{U} \rightarrow [0, 1]$.

$Q ::= \text{SEM}(\text{text} \mid \text{image})$	semantic similarity
$\mid \text{CHG}(\text{type}, \Delta t, \alpha)$	calibrated change evidence
$\mid \text{SPA}(\text{relation}, \text{layer}, \text{param})$	symbolic spatial predicate
$\mid \text{META}(\text{field}, \text{predicate})$	AOI / date / sensor filter (R1.4)
$\mid \text{PHEN}(\text{exemplar})$	phenological-regime similarity
$\mid Q \wedge Q \mid Q \vee Q \mid \neg Q$	composition

(24)

with fuzzy-logic semantics on the scored sets: $(Q_1 \wedge Q_2)(u) = Q_1(u)^\lambda Q_2(u)^{1-\lambda}$ (weighted geometric mean, so both must be satisfied), $(Q_1 \vee Q_2)(u) = \max$, $(\neg Q)(u) = 1 - Q(u)$. The parameter λ is exposed to the analyst as a single slider, generalising the modality-control idea from composed image retrieval. META and SPA are hard predicates, realised as $\{0, 1\}$ -valued sets and pushed down into the SQL **WHERE** clause for efficiency.

The example query compiles to:

$$\text{SEM}(\text{“built structures”}) \wedge \text{CHG}(\text{construction}, [t_a, t_b], \alpha) \wedge \text{SPA}(\text{ST_DWithin}, \text{hydro}, 500 \text{ m}) \quad (25)$$

5.9.3 Joint retrieval–change ranking

Requirement R1.5 and R2.1 are not separable, and we do not separate them. The final score is

$$\boxed{\text{score}(i, t) = \underbrace{\tilde{\sigma}_{\text{sem}}(q, i)^\lambda}_{\text{semantic}} \cdot \underbrace{(1 - p_i(t))^{1-\lambda}}_{\text{calibrated change}} \cdot \underbrace{\mathbb{K}[\Pi_{\text{spa}}(i)] \mathbb{K}[\Pi_{\text{meta}}(i, t)]}_{\text{hard predicates}}} \quad (26)$$

where $\tilde{\sigma}_{\text{sem}}$ is the rank-normalised semantic similarity. Setting $\lambda = 1$ recovers pure semantic retrieval; $\lambda = 0$ recovers a pure change queue; intermediate values answer the compound queries the problem statement actually poses. Because $p_i(t)$ is a calibrated p-value rather than an uncalibrated distance, the two factors are on commensurate scales — which is the reason naive score fusion between a retrieval system and a change detector normally fails.

6 System Architecture

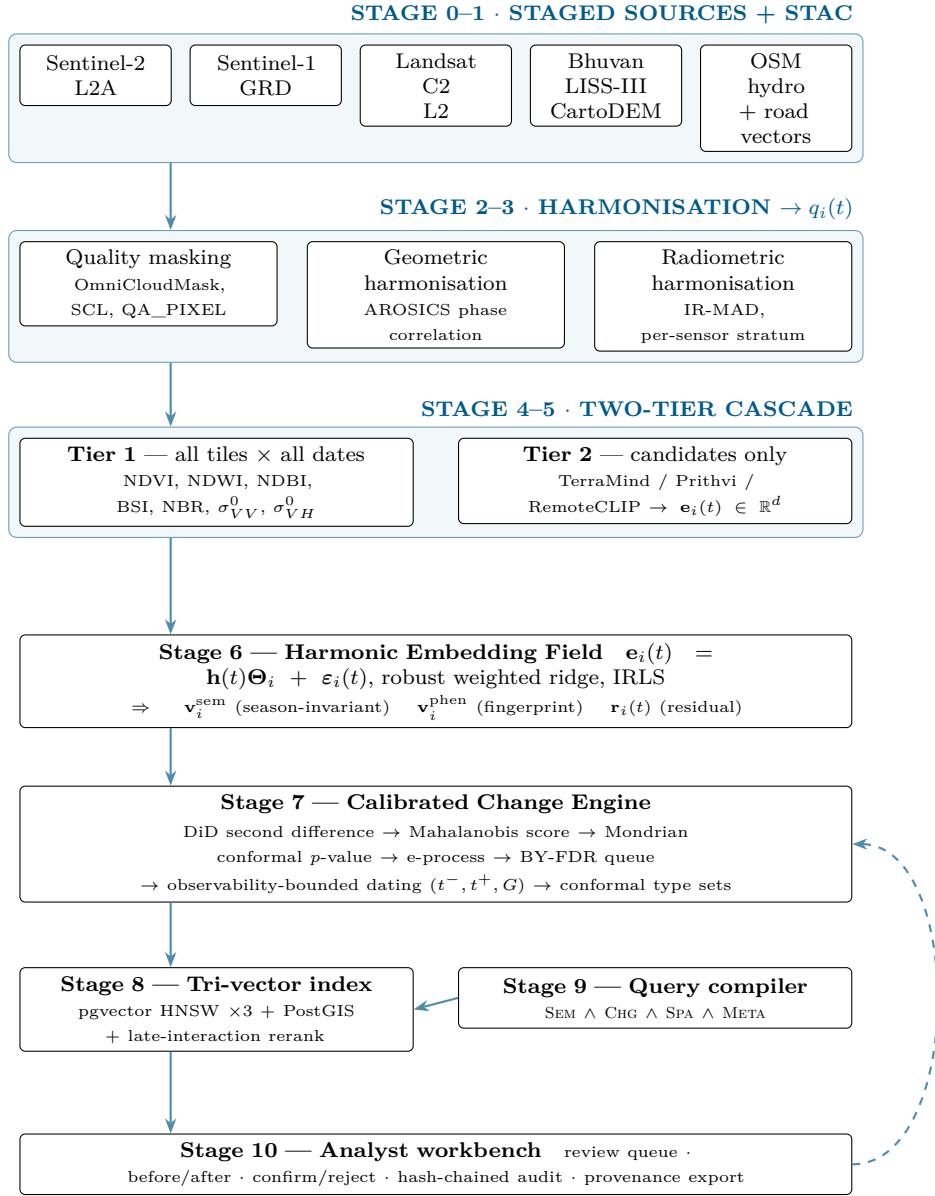


Figure 1: CHRONOS end-to-end architecture. The dashed return path is the feedback loop of §7: analyst rejections re-enter the conformal calibration set, so review *tightens the statistical guarantee* rather than merely reordering results.

6.1 Technology stack

Every component is selected under the constraint that it must run air-gapped (R6.3, R7.1) on commodity hardware.

Layer	Choice	Rationale
Raster I/O	GDAL, rasterio, rio-cogeo	COG/GeoTIFF read & write; windowed reads avoid full-scene decode (R6.5).
Catalogue	STAC 1.0 + <code>stac-fastapi-pgstac</code>	Community standard for georeferencing + acquisition metadata; local deployment, no external API (R6.4).
SAR preprocessing	ESA SNAP (graph processing) or <code>pyroSAR</code>	Orbit, border/thermal noise, calibration, terrain flattening, speckle.
Cloud masking	OmniCloudMask (ensemble U-Net) + SCL/QA_PIXEL prefilter	Current SOTA; sensor-agnostic via dynamic z -score normalisation and mixed-resolution training; needs only R/G/NIR at ≤ 50 m, so one model covers S2, Landsat and LISS-III.
Co-registration	AROSICS (phase correlation + MSSIM filtering + RANSAC)	Reported RMSE 4.45 m for L8→S2; robust across >100-day gaps; open source, offline.
Encoders	TerraMind-1.0 (Apache-2.0), Prithvi-EO-2.0 (Apache-2.0), Clay v1.5, RemoteCLIP / GeoRSCLIP, CLOSP	Multispectral + SAR + text. All weights downloadable and cacheable.
Inference	PyTorch + TerraTorch; ONNX Runtime CPU fallback	<code>BACKBONE_REGISTRY</code> loads from local cache with no network access.
Store	PostgreSQL 16 + PostGIS 3.4 + pgvector (HNSW)	Single store for geometry, metadata, vectors and audit trail. Filters in R1.4 are spatial/relational; search is vector; ACID for R5.2. Splitting these means reimplementing filtered search.
Scale fallback	FAISS (HNSW / IVF-PQ)	If tile count exceeds pgvector’s comfortable range ($> 10^7$), move vectors to FAISS and keep predicates in PostGIS.
Columnar analytics	GeoParquet + DuckDB (spatial ext.)	Major TOM’s convention; efficient bulk scans for evaluation and for harmonic fitting.
API	FastAPI + Pydantic	Typed request/response; OpenAPI for the deliverable.
UI	React + MapLibre GL JS + deck.gl	Basemap served from local PMTiles/MBTiles. No CDN fonts, no CDN JS, no remote tiles — the classic air-gap failure.
Query parsing	Local quantised LLM (llama.cpp GGUF) with constrained grammar; deterministic rule-based fallback	Offline; grammar-constrained decoding guarantees a parseable plan; the fallback means a parse failure cannot break the demo.
Orchestration	Docker Compose, images pre-pulled and <code>docker saved</code>	Reproducible bring-up with zero network.
Reproducibility	DVC or content-hash manifest; <code>pip download</code> wheelhouse	Every artefact hashed; environment reconstructible offline (R8.5).

Table 6: On-premises technology stack.

6.2 Model provenance and licensing (R7.2, R8.4)

Artefact	Origin	Licence	Offline packaging
TerraMind-1.0-{S,B,L}	IBM + ESA + FZJ	Apache-2.0	HF weights .pt, SHA-256 pinned
Prithvi-EO-2.0-{300M,600M}	IBM + NASA + JSC	Apache-2.0	HF weights, SHA-256 pinned
Clay v1.5	Clay Foundation (nonprofit)	Open source — <i>verify text</i>	HF weights
RemoteCLIP / GeoRSCLIP	Liu et al. / Zhang et al.	<i>verify per-repo</i>	GitHub/HF checkpoints
CLOSP	Rege Cambrin et al.	<i>verify per-repo</i>	HF CLOSP-Visual
OmniCloudMask v4	Wright et al.	open source	PyPI + weights
Sentinel-1/2 imagery	ESA Copernicus	Free and open (Copernicus)	staged locally
Landsat Collection 2	USGS	Public domain	staged locally
Bhuvan AWiFS/LISS-III/CartoDEM	NRSC/ISRO	NDSAP	staged locally
OSM hydro/road vectors	OpenStreetMap	ODbL (attribution)	Geofabrik extract
PostGIS	OSGeo	GPL-2.0+	container image
pgvector	Andrew Kane	PostgreSQL licence	container image

Table 7: Model and data provenance. Entries marked *verify* must have their licence text read and recorded before the submission; we do not assert a licence we have not read.

7 Implementation Stages

The system is specified as thirteen stages. Each is independently testable and produces a durable, hashed artefact.

Stage 0 — Offline Staging, Licensing and Provenance Baseline

Satisfies: R6.3, R7.1, R7.2, R7.3, R8.4.

Inputs. Network access (this is the only stage that has it).

Outputs. A sealed, hashed, self-contained deployment bundle.

1. **Imagery staging.** Download the AOI/time-span scenes from Copernicus Data Space, USGS EarthExplorer / M2M, and Bhuvan NOEDA. Convert everything to COG (`rio-cogeo`) with internal overviews and tiled layout.
2. **Auxiliary layers.** CartoDEM (or Copernicus DEM 30 m as backup), OSM extract for India (Geofabrik) filtered to `waterway=*`, `natural=water`, `highway=*`, `landuse=*`; loaded into PostGIS with GiST indices.
3. **Basemap tiles.** Render or download an offline PMTiles basemap for the AOI. *This step is the single most commonly forgotten air-gap dependency and it fails visibly, on stage.*
4. **Model weights.** Pull each checkpoint in Table 7, record SHA-256, record source URL, record licence text verbatim into `PROVENANCE.md`.

5. **Python environment.** `pip` download a complete wheelhouse; pin with a lockfile; verify that installation succeeds with `-no-index`.
6. **Container images.** Build, then `docker save` every image to tar.
7. **Seal and verify.** Compute a manifest of content hashes. **Then physically disable the network and run the full bring-up plus a smoke test.** Any step that reaches out will fail here, in rehearsal, rather than in front of the evaluators.

Acceptance test. With the network interface down, `docker compose up` succeeds, the UI loads with a visible basemap, a text query returns results, and a change query returns a review queue.

Stage 1 — Ingestion, Cataloguing and Tiling

Satisfies: R6.4, R6.5.

1. **Scene registration.** For each COG, parse CRS, geotransform, footprint, acquisition datetime, platform, instrument, processing baseline, sun/view geometry. Write a STAC Item with the `eo`, `view`, `sat` and `proj` extensions. *Nothing is discarded:* R6.4 requires that georeferencing and acquisition metadata survive to the export, so the STAC Item is the canonical record and every downstream artefact carries its ID.
2. **Tile grid.** Define a fixed, sensor-independent analysis grid in the AOI's UTM zone. Cell size $P = 224$ px at the reference GSD $\rho_0 = 10$ m $\Rightarrow$ 2.24 km, with 50% overlap (stride 1.12 km). Overlap raises recall for objects straddling boundaries and gives each location two independent looks, which we exploit as a consistency check.
3. **Rationale for $P = 224$.** It matches the native input of the ViT encoders (avoiding resampling artefacts), and with patch size 16 it yields a 14×14 token grid at 160 m — the resolution at which late-interaction retrieval localises evidence within a tile (§7).
4. **Resolution policy.** Coarser sensors are *not* upsampled into the 10 m grid for feature extraction (that fabricates detail). Instead each sensor keeps its native grid, and tiles are related by the common geometry g_i . Landsat contributes a 75×75 px view of the same 2.24 km footprint; LISS-III a 95×95 px view.
5. **Multi-scale option.** A coarse level L−1 at 8.96 km supports wide-area context queries and dramatically accelerates AOI-scale screening.
6. **Storage.** Tile-level records go to GeoParquet partitioned by (MGRS cell, year); geometry, IDs and vectors go to PostGIS/pgvector.

Acceptance test. Round-trip: pick a random tile ID, recover its exact geometry, its source scene, and re-extract the identical pixels from the COG.

Stage 2 — Quality Masking and Geometric Harmonisation

Satisfies: R3.1, R3.2.

This stage produces $q_i(t)$, the quality weight that propagates into *every* downstream estimator (the harmonic weights in Eq. 4, the e-process gating, and the observability gap G). It is therefore load-bearing.

2a. Cloud, shadow and snow.

- Cheap prefilter: Sentinel-2 SCL and Landsat QA_PIXEL. Fast, free with the product, but conservative — it systematically under-detects thin cloud and haze.
- Primary mask: OmniCloudMask, an ensemble of two U-Nets with soft voting, requiring only red/green/NIR at ≤ 50 m. Because it is sensor-agnostic by design, *one* model and one set of weights covers Sentinel-2, Landsat and LISS-III — a material simplification of the offline bundle.
- Snow: NDSI threshold plus SCL snow class. Snow is treated as a quality reduction, not as change, but a *persistent* anomalous snow state is genuine and is preserved by the harmonic model (snow is seasonal; anomalous snow produces residual).

2b. Haze. Even after L2A atmospheric correction, residual aerosol is a leading cause of false change. We compute a per-tile haze index from the blue/aerosol band and fold it into q .

2c. Geometric co-registration. This addresses the 1.2–2.8 px multi-temporal misregistration measured for Sentinel-2 [5] and the ≈ 38 m cross-sensor error.

1. Build a per-tile *reference*: the temporal median composite over all high- q observations. A median composite is cloud-free by construction and is the most stable available registration target.
2. Estimate a sub-pixel shift $(\delta x, \delta y)$ per observation by phase correlation (AROSICS): dense tie points on a regular grid, outlier rejection by mean structural similarity, affine model by RANSAC.
3. Resample by the estimated shift (Lanczos), *or* — our preference where the shift is small — leave the pixels alone and record the shift.
4. **Feed the registration residual into q .** This is a design decision we consider important: rather than only correcting misregistration, we *measure* it and let a poorly registered observation carry less weight in the harmonic fit and contribute a weaker e-value. Registration uncertainty thus propagates into the final confidence rather than silently contaminating it.

2d. SAR-specific quality. Layover and radar-shadow masks derived from CartoDEM; local incidence angle recorded and used as a stratification variable (a change in σ^0 that correlates with incidence-angle change across the cohort is geometric, not physical).

Composite weight.

$$q_i(t) = \underbrace{(1 - \text{cloud}_i(t))}_{\text{clear fraction}} \cdot \underbrace{(1 - \text{shadow}_i(t))}_{\text{shadow}} \cdot \underbrace{e^{-\text{haze}_i(t)/h_0}}_{\text{haze}} \cdot \underbrace{e^{-\|\delta\|/\delta_0}}_{\text{registration}} \cdot \underbrace{(1 - \text{layover}_i)}_{\text{SAR only}}. \quad (27)$$

Observations with $q_i(t) < q_{\min}$ (default 0.5) are excluded from fitting but *retained in the record*, because they are exactly what populates the observability gap G in R2.3.

Stage 3 — Radiometric Harmonisation and Cross-Sensor Alignment

Satisfies: R3.1, R3.2.

1. **Base products.** Sentinel-2 L2A BOA and Landsat C2 L2 surface reflectance are already atmospherically corrected — the largest part of the job is done upstream.

2. **Within-sensor relative normalisation.** IR-MAD (iteratively reweighted multivariate alteration detection) between each observation and the tile reference composite. IR-MAD automatically identifies invariant pixels statistically, avoiding the known fragility of manual pseudo-invariant-feature selection in rural, vegetation-dominated scenes where built PIFs are scarce.
3. **Cross-sensor band mapping.** Landsat OLI $\rightarrow$ Sentinel-2 MSI band correspondence with published linear coefficients (HLS-style). LISS-III’s four bands map to S2 B3/B4/B8/B11.
4. **BRDF / view-angle.** A c-factor BRDF normalisation to nadir, using CartoDEM for slope and aspect. This addresses “view-angle differences” in R3.1 directly.
5. **Residual bias handled by stratification, not correction.** Per the remark in §3: whatever cross-sensor bias survives steps 2–4 is absorbed by including sensor in the Mondrian stratum κ . The conformal threshold for a Landsat-observation-in-winter-over-cropland is calibrated on Landsat-in-winter-over-cropland nulls. This is more honest and more robust than pretending a perfect harmonisation exists.

Stage 4 — Tier-1 Physical Feature Time Series

Satisfies: R2.2, R3.3; enables the cascade that makes the compute budget feasible.

Computed for **all tiles at all dates**. Pure array arithmetic — milliseconds per tile, no GPU.

Feature	Definition	Diagnostic for
NDVI	$(\text{NIR} - \text{R})/(\text{NIR} + \text{R})$	vegetation, clearance
NDWI	$(\text{G} - \text{NIR})/(\text{G} + \text{NIR})$	open water
MNDWI	$(\text{G} - \text{SWIR}_1)/(\text{G} + \text{SWIR}_1)$	water vs. built confusion
NDBI	$(\text{SWIR}_1 - \text{NIR})/(\text{SWIR}_1 + \text{NIR})$	built-up
BSI	bare soil index	clearance, site preparation
NBR	$(\text{NIR} - \text{SWIR}_2)/(\text{NIR} + \text{SWIR}_2)$	burn, severe disturbance
TC brightness/greenness/wetness	tasselled cap	general
$\sigma_{VV}^0, \sigma_{VH}^0, \text{VH/VV}$	SAR backscatter	structure, water, moisture
Textural (GLCM contrast, homogeneity)	grey-level co-occurrence	structural change

Each feature gets its own harmonic fit (Stage 6 applies identically, with $d = 1$). The Tier-1 residual vector is the *screening* statistic: tiles whose Tier-1 e-process exceeds a permissive threshold become Tier-2 candidates.

Remark 17 (Why the cascade is not a compromise). Cheap physical screening followed by expensive semantic confirmation is not merely a compute optimisation. It is the precision-oriented architecture that R3.3 requires: a candidate must satisfy both a physical criterion and a semantic criterion. It also makes the build-time and storage figures in the mandatory evaluation report (R8.5) defensible rather than embarrassing.

Stage 5 — Tier-2 Semantic Embedding Extraction

Satisfies: R1.1, R1.2, R4.1.

1. **Encoder selection.** GEO-Bench-2 establishes that no single model dominates [12], so we run a small, principled ensemble rather than betting on one:
 - **TerraMind-1.0-B** — multispectral + SAR, leads PANGAEA, Apache-2.0.
 - **Prithvi-EO-2.0-300M-TL** — multi-temporal with temporal and location embeddings; strongest on multispectral tasks.

- **RemoteCLIP / GeoRSCLIP** — required for text-to-image (R1.1); the image-only GFMs have no text tower.
 - **CLOSP** — if staged, gives a unified SAR + multispectral + text space, the only credible route to text-searching SAR.
2. **Which vector for which job.** Text retrieval must use a VLM (only it has an aligned text tower). Change detection should use a GFM (better multispectral and temporal representation). We therefore maintain *parallel* embedding channels and let the query type route.
 3. **Sampling policy.** Not every tile at every date. We embed (a) a stratified seasonal sample — roughly 12–16 dates per tile spanning all phases, sufficient to fit $K = 2$ with margin, and (b) every date for Tier-1 candidate tiles, and (c) every date within ± 3 acquisitions of a Tier-1 candidate break.
 4. **Multi-scale.** Embed at L0 (2.24 km tiles) and, for candidate tiles only, at L1 (560 m sub-tiles, i.e. a 4×4 subdivision) for finer change localisation.
 5. **Normalisation.** ℓ_2 -normalise before all downstream use, so cosine and Euclidean geometry agree and the Mahalanobis covariance is well-conditioned.
 6. **Compression.** Matryoshka truncation to $d' \in \{256, 512\}$ plus int8 scalar quantisation for the stored index, retaining full precision only for the candidates being reranked. Published results retain $\approx 96\%$ of FP32 nDCG@10 at 192 dimensions with aggressive quantisation.

Stage 6 — Harmonic Embedding Field Fitting

Satisfies: R3.1 (season), R2.3, R4.1; theory in §5.1.

Algorithm 1 Fit Harmonic Embedding Field for tile i

Require: observations $\{(t_j, \mathbf{e}_i(t_j), q_i(t_j))\}_{j=1}^{n_i}$, order K , ridge λ

Ensure: $\hat{\Theta}_i$, residuals $\mathbf{r}_i(\cdot)$, fit diagnostics

```

1: if  $n_i < 3(2 + 2K)$  or temporal span  $< 1.5$  yr then
2:   return FALLBACKFIT( $\cdot$ ) ▷ §7.1
3: end if
4:  $\mathbf{H} \leftarrow [\mathbf{h}(t_1); \dots; \mathbf{h}(t_{n_i})]$ ;  $\mathbf{W} \leftarrow \text{diag}(q_i(t_j))$ 
5:  $\hat{\Theta} \leftarrow (\mathbf{H}^\top \mathbf{W} \mathbf{H} + \lambda \mathbf{\Lambda}^\top \mathbf{\Lambda})^{-1} \mathbf{H}^\top \mathbf{W} \mathbf{E}$  ▷ Eq. 5
6: for  $\ell = 1$  to  $L_{\max}$  do ▷ IRLS,  $L_{\max} = 5$ 
7:    $\mathbf{r}(t_j) \leftarrow \mathbf{e}_i(t_j) - \mathbf{h}(t_j) \hat{\Theta}$ 
8:    $\hat{\varsigma} \leftarrow 1.4826 \cdot \text{MAD}(\{\|\mathbf{r}(t_j)\|\})$ 
9:    $w_j \leftarrow q_i(t_j) \cdot \min(1, \delta \hat{\varsigma} / \|\mathbf{r}(t_j)\|)$ 
10:  recompute  $\hat{\Theta}$  with  $\mathbf{W} = \text{diag}(w_j)$ 
11:  break if  $\|\hat{\Theta}^{(\ell)} - \hat{\Theta}^{(\ell-1)}\|_F < \epsilon$ 
12: end for
13:  $\mathbf{v}_i^{\text{sem}} \leftarrow \hat{\alpha}_i + \hat{\beta}_i \bar{t}_{\text{mid}}$ 
14:  $\mathbf{v}_i^{\text{phen}} \leftarrow \text{AMPLITUDEPHASE}(\hat{\gamma}_i, \hat{\delta}_i)$ 
15: cache  $\mathbf{P}_i \leftarrow (\mathbf{H}^\top \mathbf{W} \mathbf{H} + \lambda \mathbf{\Lambda}^\top \mathbf{\Lambda})^{-1}$  ▷ for recursive update, Stage 11
16: return  $\hat{\Theta}_i$ ,  $\mathbf{r}_i$ ,  $\mathbf{v}_i^{\text{sem}}$ ,  $\mathbf{v}_i^{\text{phen}}$ ,  $\mathbf{P}_i$ , GCV score,  $n_i^{\text{eff}}$ 
```

Diagnostics stored per tile (and surfaced in the UI). Effective sample size $n^{\text{eff}} = (\sum w)^2 / \sum w^2$; temporal span; largest observation gap; GCV score; fitted seasonal amplitude A_1, A_2 ; residual scale. These are what let an analyst judge *how much to trust* a given detection — provenance in the sense of R5.1 extends to the statistical model, not just the pixels.

Cost. With $m = 6$, per tile the fit is $O(n_i m^2 + m^3 + n_i m d)$. For $n_i = 77$, $d = 768$: roughly 4×10^5 FLOPs, i.e. microseconds. **The harmonic layer is computationally free relative to the embedding extraction** — an important practical point, since the intellectual centrepiece of the system costs almost nothing to run.

Stage 7 — Calibrated Change Engine

Satisfies: R2.1, R2.2, R2.3, R3.1, R3.2, R3.3; theory in §§5.3–5.8.

Algorithm 2 Calibrated change decision for tile i at time t

Require: residuals $\mathbf{r}_i(t)$ for the acquisition, cohort index, calibration store, α , q

- 1: $\mathcal{C}(i) \leftarrow k\text{-NN of } \mathbf{v}_i^{\text{phen}} \text{ within same acquisition, elevation and aspect band}$
- 2: $\tilde{\mathbf{r}}_i(t) \leftarrow \mathbf{r}_i(t) - \text{med}_{m \in \mathcal{C}(i)} \mathbf{r}_m(t)$ ▷ DiD, Eq. 10
- 3: $\rho \leftarrow \text{cohort inlier fraction}$ ▷ Assumption 2 diagnostic
- 4: **if** $\rho < \rho_{\min}$ **then**
- 5: route to REGIONALEVENTQUEUE; **return**
- 6: **end if**
- 7: $\kappa \leftarrow (\ell_i, \text{season}(t), \text{sensor}(t), \rho_i)$
- 8: $s_i(t) \leftarrow \sqrt{\tilde{\mathbf{r}}_i(t)^\top \hat{\Sigma}_\kappa^{-1} \tilde{\mathbf{r}}_i(t)}$
- 9: $p_i(t) \leftarrow (1 + |\{j \in \mathcal{D}_{\text{cal}}^\kappa : s_j \geq s_i(t)\}|) / (|\mathcal{D}_{\text{cal}}^\kappa| + 1)$ ▷ Thm. 12
- 10: **if** $q_i(t) \geq q_{\min}$ **then**
- 11: $E_i \leftarrow E_i \cdot \varkappa p_i(t)^{\varkappa-1}$ ▷ e-process, Thm. 15
- 12: **end if**
- 13: **if** $E_i \geq 1/\alpha$ **then**
- 14: $(t^-, t^+, G) \leftarrow \text{OBSERVABILITYBOUNDS}(i)$ ▷ §5.6
- 15: $\hat{Y}_i \leftarrow \text{CONFORMALTYPESET}(\Delta_i)$ ▷ §5.8
- 16: emit candidate with $(p, E, t^-, t^+, G, \hat{Y}, \rho, n^{\text{eff}}, \text{lineage})$
- 17: **end if**
- 18: **periodically:** apply Benjamini–Yekutieli at level q over the candidate pool $\rightarrow$ review queue

Change direction (R2.1). On confirmation, the tile-level decision is handed to the change-morphology module of §5.7, which mosaics the patch-resolution residual field across the 50% tile overlap, extracts connected components as polygonal change *objects*, matches them across epochs by IoU, and classifies appearance / disappearance / expansion / contraction from the match structure and a conformally calibrated relative-area statistic. Objects that split or merge are reported as their own class rather than coerced into the four. The exported geometry is the object polygon, not the tile.

Confidence reported to the analyst. Not a raw score. Three numbers with meanings: $p_i(t)$ (calibrated per-decision false-alarm probability), E_i (accumulated anytime-valid evidence), and the BY-adjusted queue-level FDR q . Plus the diagnostic triple $(\rho, n^{\text{eff}}, \text{span})$ that says how much to trust the underlying model fit.

Stage 8 — Index Construction, Retrieval and Discovery

Satisfies: R1.1–R1.4, R4.1, R4.2, R6.1, R6.2.

8a. The tri-vector index.

Index	Content	Dim	Serves
<code>idx_sem</code>	$\mathbf{v}_i^{\text{sem}}$, season-invariant	$d' = 256\text{--}512$ (int8)	R1.1, R1.2
<code>idx_phen</code>	$\mathbf{v}_i^{\text{phen}}$, amplitude/phase	$2Kd'$ or PCA-reduced	R4.1, R4.2
<code>idx_res</code>	$\tilde{\mathbf{r}}_i(t)$, sparse (stored only where $p < 0.2$)	d'	R2.1

All three are pgvector HNSW indices in the same database as the geometry and metadata — so R1.4’s AOI/date/sensor filters are ordinary SQL predicates rather than a second system.

8b. Filtered ANN strategy. Filter selectivity determines the plan:

- *High selectivity* (small AOI, narrow date range): pre-filter in PostGIS via the GiST index, then exact search over the surviving vectors.
- *Low selectivity* (archive-wide query): HNSW search with post-filtering and progressive k' doubling until k qualified results are found.
- *Middle regime*: ACORN-style predicate-agnostic traversal with two-hop expansion, which avoids the pathologies of both extremes.

8c. Late-interaction reranking. Two-stage:

1. Retrieve top- $K_1 \approx 1000$ by single-vector ANN over `idx_sem`.
2. Apply hard predicates.
3. Rerank top- $K_2 \approx 200$ by MaxSim over patch tokens,

$$\text{MaxSim}(q, i) = \sum_{u \in q} \max_{v \in \text{patches}(i)} \langle \mathbf{z}_u, \mathbf{z}_v \rangle,$$

with patch tokens *recomputed on demand* rather than stored.

This buys the documented late-interaction accuracy gain without the documented storage cost, and it produces a per-patch relevance heatmap at 160 m — the “where in this tile” answer the analyst actually wants. Where storage permits, HPC-ColPali-style centroid quantisation ($\approx 32\times$ reduction) plus attention-guided pruning makes persistence viable.

8d. Discovery and clustering (R4.1, R4.2).

- **Query-by-example:** k -NN from a seed tile over $[\zeta \mathbf{v}^{\text{sem}} \parallel (1 - \zeta) \mathbf{v}^{\text{phen}}]$. The ζ slider trades “looks like this” against “behaves like this through the year” — and the second is frequently what an analyst means by “similar site”.
- **Unsupervised grouping:** HDBSCAN over the same concatenation, yielding variable-density clusters with an explicit noise class (unlike k -means, which forces every tile into a cluster). Cluster exemplars and a phenological profile plot are surfaced per cluster.
- **Why this beats single-date clustering:** two sites that look identical in a single August scene — an irrigated double-cropped field and a permanently vegetated plantation — have entirely different $\mathbf{v}^{\text{phen}}$ and separate cleanly. This capability is a direct consequence of the trajectory representation and is not available to any point-based system.

Stage 9 — Query Compilation and Serving

Satisfies: R1.1, R1.4, R1.5; theory in §5.9.2.

1. **Parse.** A local quantised LLM with grammar-constrained decoding emits a query plan in the algebra of §5.9.2. Grammar constraint guarantees syntactic validity. A deterministic keyword/pattern parser is the fallback — *the demo must not be able to fail on a parse error*.
2. **Ground.** Map spatial nouns to staged vector layers via a curated lexicon (`river/stream/canal` → `waterway`; `road/highway` → `highway`; `settlement/town` → `landuse=residential`). Map relations to PostGIS operators with default parameters (`near` → `ST_DWithin`, 500 m default, analyst-adjustable; `within` → `ST_Within`; `along` → `ST_DWithin` + elongation constraint).
3. **Plan and optimise.** Push hard predicates down into SQL; estimate selectivity; choose the ANN strategy per 8b.
4. **Execute and fuse.** Score per §5.9; rerank per 8c.
5. **Explain.** Return the compiled plan alongside the results. The analyst sees exactly which operator produced which constraint — interpretability is a defence requirement, not a nicety, and a black-box ranking is not auditable.

Stage 10 — Analyst Workflow, Provenance and Feedback

Satisfies: R5.1–R5.4.

10a. Review queue (R5.1). Each item presents: before/after chips at t^- and t^+ with a swipe comparator; the residual-magnitude heatmap overlaid; the location (coordinates, and nearest named place from OSM); acquisition times and platforms for both chips; the calibrated confidence triple; the conformal type set $\hat{Y}_i$; the full time series plot with the fitted harmonic curve, the observations, the quality weights and the marked break; and an expandable processing-history panel.

10b. The time-series plot is the killer evidence panel. It is worth stating why. A before/after pair is what every competing system shows, and it is exactly the display that *cannot* distinguish season from change. Our panel shows the analyst the location’s entire history, the model of its normal seasonal behaviour, and the point at which it departed from that model — which is a self-evidently more convincing argument, and one only a trajectory-based architecture can make.

10c. Audit trail (R5.2). Append-only PostgreSQL table; each row contains (`candidate_id`, `analyst_id`, `decision`, `rationale`, `timestamp`, `prev_hash`) where `prev_hash` is the SHA-256 of the previous row’s canonical serialisation. The result is a tamper-evident hash chain: any retrospective alteration invalidates every subsequent hash. Rows are never updated or deleted; a reversal is a new row.

10d. Feedback (R5.3). Three mechanisms, in increasing order of interest:

1. *Query-point movement* (Rocchio) in embedding space: $\mathbf{q}' = a\mathbf{q} + \frac{b}{|R|} \sum_{i \in R} \mathbf{v}_i - \frac{c}{|R|} \sum_{i \in \bar{R}} \mathbf{v}_i$. Immediate, no training.
2. *Lightweight reranker*: logistic regression over $[\tilde{\sigma}_{\text{sem}}, p, E, \Delta \text{indices}, \text{metadata}]$. CPU-only, seconds to fit, and — importantly for a defence context — its coefficients are directly inspectable, so the analyst can see *why* the ranking moved.

3. **Rejections become conformal nulls.** An analyst rejection is a human-certified $\mathcal{H}_0$ sample — precisely what Theorem 12 needs. Rejections are appended to $\mathcal{D}_{\text{cal}}^\kappa$ for their stratum, tightening the threshold exactly where the system is making mistakes. **Review therefore improves the statistical guarantee itself, not merely the ranking.** We are not aware of any prior system in which the human-in-the-loop and the formal guarantee reinforce each other in this way, and it is a natural consequence of having a guarantee at all.

10e. Export (R5.4). GeoPackage / GeoJSON plus STAC Items, each record carrying: source scene IDs for both epochs; the full processing DAG (algorithm, version, parameters); model names, versions and weight SHA-256; the conformal α and BY q in force; the diagnostic triple; the analyst decision and its audit-chain hash; and CRS and geometry. An export is reproducible and attributable end to end.

Stage 11 — Incremental Ingestion

Satisfies: R6.2 — “incremental addition of newly acquired imagery without a complete index rebuild.”

Every stateful component of CHRONOS updates in closed form. This is not an accident of engineering; it is a consequence of having chosen linear-in-parameters models throughout.

1. **Harmonic model — recursive least squares.** Maintain $\mathbf{P}_i = (\mathbf{H}^\top \mathbf{W} \mathbf{H} + \lambda \mathbf{\Lambda}^\top \mathbf{\Lambda})^{-1}$. On a new observation $(\mathbf{h}_*, \mathbf{e}_*, w_*)$, apply Sherman–Morrison:

$$\mathbf{k} = \frac{w_* \mathbf{P}_i \mathbf{h}_*^\top}{1 + w_* \mathbf{h}_*^\top \mathbf{P}_i \mathbf{h}_*}, \quad \hat{\boldsymbol{\Theta}}_i \leftarrow \hat{\boldsymbol{\Theta}}_i + \mathbf{k}(\mathbf{e}_* - \mathbf{h}_*^\top \hat{\boldsymbol{\Theta}}_i), \quad \mathbf{P}_i \leftarrow \mathbf{P}_i - \mathbf{k} \mathbf{h}_* \mathbf{P}_i. \quad (28)$$

Cost $O(m^2 + md) = O(36 + 6d)$ per tile. *No refit, no rebuild.* This is the same recursive structure that S-CCD uses for near-real-time monitoring, generalised to d dimensions.

2. **Covariance — rank-one update** of $\hat{\boldsymbol{\Sigma}}_\kappa$ with exponential forgetting, so the noise model tracks slow sensor drift.
3. **Vector index — HNSW insertion.** $O(\log N)$ per vector, natively supported. This is why HNSW and not a flat IVF index, which would require retraining the coarse quantiser.
4. **E-process — multiplicative update.** $E_i \leftarrow E_i \cdot \varsigma_\kappa(p_{\text{new}})$. By Theorem 15 the guarantee is preserved *with no recalibration and no multiple-testing correction* — the single most important practical consequence of choosing an anytime-valid procedure.
5. **Conformal calibration — streaming quantile.** Maintain a bounded reservoir per stratum with a t-digest for the $(1 - \alpha)$ quantile; optionally run adaptive conformal inference, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathbb{I}[\text{err}_t])$, to hold empirical coverage under drift.
6. **Control cohorts — lazily rebuilt** from `idx_phen`, which is itself incrementally maintained.
7. **STAC catalogue — append** a new Item; all downstream artefacts reference it by ID.

Complexity of ingesting one new scene: $O(N_{\text{scene}} \cdot (\text{mask} + \text{coreg} + \text{Tier-1}))$ plus $O(N_{\text{cand}} \cdot \text{embed})$ plus $O(N_{\text{scene}}(m^2 + md + \log N))$. *Linear in the new data, logarithmic in the archive.* Nothing scales with total archive size except the HNSW insertion, and that logarithmically.

Stage 12 — Evaluation Harness and Reporting

Satisfies: R3.3, R8.5. Detailed protocol in §9.

Instrumented from day one, not bolted on: every stage emits structured timing, memory and artefact-size telemetry to a metrics table, and the evaluation report is generated from that table by script. A reproducible report is a deliverable (R8.5), and reports assembled by hand the night before are neither reproducible nor accurate.

7.1 Fallback paths

Honest engineering requires that each contribution degrade gracefully.

Condition	Consequence	Fallback
Time span < 1.5 yr	Harmonic model unidentifiable	Not a fallback but a designed operating mode — see §5.2 and Table 4. Mode B ($K = 1$, transferred seasonal prior) or Mode C (bi-temporal), selected per tile. C2 and C3 operate unchanged in all three modes, so the calibrated guarantee survives.
$n_i < 3m$ usable obs	Overfitting risk	Increase λ ; pool with stratum-level prior (hierarchical shrinkage); flag reduced n^{eff} in the UI.
Persistent cloud (monsoon)	Optical series collapses	SAR carries the series; the observability gap G is reported honestly rather than a false date being produced.
Sparse stratum κ	Conformal threshold unreliable	Hierarchical pooling to parent stratum; report effective calibration n ; widen the interval.
Cohort contaminated ($\rho < \rho_{\min}$)	Assumption 2 violated	Route to the regional-event queue with the first-stage residual $\mathbf{r}$ surfaced.
Query parse failure	No plan	Deterministic rule-based parser; ultimately pure semantic search.
Encoder unavailable / OOM	No Tier-2	Tier-1 physical features alone still deliver a functioning, calibrated change engine — the whole statistical apparatus is encoder-agnostic.

Table 8: Degradation paths. Note the last row: because the conformal and DiD machinery operates on *any* residual series, the system retains its central guarantee even with no foundation model at all.

8 Complexity and Resource Analysis

8.1 Asymptotics

8.2 Worked example

AOI 200×200 km ($40,000$ km², comparable to a large Indian district cluster), three years, Sentinel-2 primary.

Remark 18. Three observations from Table 10. First, the cascade reduces Tier-2 forward passes from 2.46×10^6 (naive: every tile, every date) to 6.8×10^5 , cutting the embedding stage from 3.4 h

Stage	Time	Space	Incremental cost
1 Tiling	$O(NTBP^2)$	$O(NTBP^2)$ streamed	$O(N_{\text{scene}}BP^2)$
2 Masking + coreg	$O(NTP^2) + \text{CNN}$	$O(NT)$	$O(N_{\text{scene}}P^2)$
4 Tier-1	$O(NTFP^2)$	$O(NTF)$	$O(N_{\text{scene}}FP^2)$
5 Tier-2 embed	$O(N_{\text{emb}} \cdot \text{FLOP}(\phi))$	$O(N_{\text{emb}}d)$	$O(N_{\text{cand}}\text{FLOP}(\phi))$
6 Harmonic fit	$O(N(nm^2 + m^3 + nmd))$	$O(Nmd)$	$O(m^2 + md)$ per obs
7 DiD + conformal	$O(NTkd + NTd^2)$	$O(\mathcal{D}_{\text{cal}})$	$O(kd + d^2)$
8 HNSW build	$O(N \log N \cdot d')$	$O(Nd')$	$O(\log N \cdot d')$ per insert
9 Query	$O(\log N + K_2 \cdot \text{MaxSim})$	—	—

Table 9: Complexity. N tiles, T timestamps, B bands, P tile side, F Tier-1 features, $m = 2 + 2K$, d embedding dim, d' compressed dim, k cohort size.

to under an hour. That is a real saving but it is not the dominant term: **this build is I/O-bound, not GPU-bound**. The binding constraints are reading a few hundred gigabytes of source imagery several times over and performing 2.46×10^6 per-tile phase correlations, which is why the honest total is 8–15 h rather than the 4–6 h a GPU-only accounting would suggest. Second, one Sentinel-2 MGRS scene covers 110×110 km, so at a 1.12 km stride it touches $(110/1.12)^2 \approx 9,600$ tiles — incremental ingestion is minutes, not seconds, and the live demonstration of R6.2 should be planned accordingly. Third, **the entire searchable index is about 1 GB** for a 40,000 km², three-year archive, three orders of magnitude below the imagery it indexes. That is the number worth putting on a slide: the semantic and temporal index of a district-scale archive fits comfortably on a laptop, which is precisely the sovereignty argument the problem statement is reaching for.

9 Evaluation Protocol

9.1 Retrieval (R1)

Metrics: nDCG@{5,10,20}, mAP, Recall@{1,5,10,50}, MRR against held-out relevance judgements. Reported as an overall figure *and* broken out by query class:

Class	Example	Hypothesis
Monadic semantic	“dense urban area”	baseline; CLIP-family competitive
Attribute	“large circular reservoir”	baseline competitive
Relational	“structures near a river”	we expect a decisive margin (§5.9)
Temporal-compound	“newly built structures near a river”	we expect a decisive margin
Image-to-image	exemplar tile	baseline competitive
Phenological	“land with this seasonal cycle”	only we can answer

Baselines: CLIP zero-shot, RemoteCLIP, GeoRSCLIP, single-vector Major TOM-style index.

9.2 Change detection (R2, R3)

Primary metrics. Precision, recall, F_1 and PR-AUC at the conformal operating point on the held-out change / no-change set.

The headline figure: the calibration curve. Plot nominal α against empirical false-alarm rate on the held-out no-change set. A valid procedure lies on or below the diagonal. This single plot is the paper’s central empirical claim and *no competing system can produce it*, because no competing system has a nominal α to plot against.

Confounder-stratified false-alarm rate. Reported separately for each nuisance category named in R3.1:

Quantity	Value	Derivation
Tile side	2.24 km	224 px $\times$ 10 m
Stride (50% overlap)	1.12 km	
Tiles N	31,900	$(200/1.12)^2$
S2 acquisitions, 3 yr	≈ 219	5-day revisit
Usable after masking ($\approx 35\%$)	≈ 77	T
Tier-1 evaluations	2.46×10^6	$N \times T$; $\sim$ ms each, CPU-parallel
Tier-2, seasonal sample	4.8×10^5	$N \times 15$ dates
Tier-2, candidates ($\sim 8\%$)	2.0×10^5	$0.08N \times T$
Total forward passes	$\approx 6.8 \times 10^5$	
Embedding wall-clock	≈ 1.0 h	at 200 img/s, single RTX 4090
Embeddings, fp32 $d=768$	2.1 GB	$6.8 \times 10^5 \times 768 \times 4$
Embeddings, int8 $d'=256$	174 MB	Matryoshka + quantisation
Harmonic coefficients	588 MB	$N \times m \times d \times 4$, $m=6$
Tier-1 series	128 MB	$N \times T \times F \times 4$, $F=13$
HNSW indices ($3\times$)	≈ 120 MB	one vector per <i>tile</i> , not per tile-date
Tile chips (COG, JPEG-compressed overviews)	≈ 45 GB	evidence display
Total index footprint (excl. imagery)	≈ 1.0 GB	
Staged source imagery	$\approx 250\text{--}400$ GB	the actual disk driver
Expected query latency (ANN + filter)	< 120 ms	p95, warm cache
Expected query latency (+ MaxSim rerank $K_2=200$)	< 450 ms	p95
Embedding wall-clock (Tier 2)	≈ 1.0 h	
Co-registration + Tier 1 + masking	4–9 h	I/O-bound; 2.46×10^6 tile registrations
Full build wall-clock	8–15 h	single workstation, dominated by I/O
Incremental ingest, one S2 scene	15–30 min	$\approx 9,600$ tiles touched

Table 10: Worked resource estimate. Target hardware: one workstation, 1 $\times$ RTX 4090 (24 GB), 64 GB RAM, 2 TB NVMe.

Stratum	Construction of the test cases	What it isolates
Seasonal	same location, same year, opposite phenological phase	$f_i(t)$
Inter-annual	same location, same phase, different years	trend, drift
Illumination	large solar-zenith difference	τ
View angle	adjacent-orbit pair, different incidence	τ , BRDF
Residual cloud/haze	moderate- q observations	masking failure
Snow	winter high-altitude pairs	$f_i(t)$, masking
Cross-sensor	S2 vs. Landsat vs. LISS-III of the same date	Stage 3, Mondrian κ
Registration	pairs with known residual shift > 1 px	Stage 2c

Gap 8 of our literature review is precisely that nobody reports this table. We report it.

Ablation ladder. Each rung must move precision at fixed recall, or the corresponding contribution is decorative and should be said to be:

#	Configuration	Isolates	Expectation
A0	Raw bi-temporal embedding distance	—	baseline
A1	+ quality masking and co-registration (Stage 2)	R3.2	moderate gain
A2	+ harmonic decomposition (C1)	season	large gain
A3	+ control-cohort DiD (C2)	acquisition effects	large gain
A4	+ Mondrian conformal threshold (C3)	threshold arbitrariness	calibration, modest P gain
A5	+ e-process persistence (C4)	transients	precision gain, small recall loss
A6	+ BY-FDR over the queue	queue quality	queue-level guarantee
A7	+ SAR fusion	cloud gaps	recall gain in monsoon strata

9.3 Change dating (R2.3)

Following the WATCH protocol so our numbers are directly comparable to published work: exact-period recall ($m = 0$) and recall within ± 1 , ± 2 , ± 3 acquisition intervals. Additionally, and uniquely: *interval coverage* — the fraction of ground-truth change dates that fall inside our reported $[t^-, t^+]$ observability interval, which should be $\geq 1 - \alpha$ if the procedure is sound, and *interval width*, since a trivially wide interval achieves coverage without being useful. Coverage and width must be reported together.

9.4 Discovery (R4)

Given a seed site, precision@ k of retrieved comparable sites against analyst judgement; cluster purity against any available land-cover reference; and an ablation of $\mathbf{v}^{\text{sem}}$ alone versus $\mathbf{v}^{\text{sem}} \parallel \mathbf{v}^{\text{phen}}$, which is the direct test of whether the phenological fingerprint carries independent information.

9.5 Systems (R8.5)

Auto-generated from Stage 12 telemetry: indexed area (km^2); scene count; tile count; embedding count; build wall-clock by stage; peak RAM and VRAM; storage footprint by artefact class; query latency p50/p95/p99 by query class; incremental ingest throughput (scenes/hour, tiles/second); and full hardware specification. Plus an offline reproducibility attestation: the manifest hash and a recorded transcript of a from-scratch, network-disabled rebuild.

10 Risks and Limitations

#	Risk	Assessment and mitigation
1	Short evaluation time span degrades the harmonic model. C1 needs ≥ 1.5 years and ≥ 18 usable observations; the held-out change set may even consist of bi-temporal pairs.	Highest-probability risk, and the reason §5.2 makes bi-temporal a first-class mode rather than a fallback. C2 and C3 — the difference-in-differences estimator and the conformal guarantee, which are the principal contributions — are independent of C1 and operate unchanged in Mode C. What is forfeited is the fingerprint, sub-annual dating and the e-process. Prototype Mode A first to establish the ceiling, but build Mode C early.

#	Risk	Assessment and mitigation
2	Harmonic fit may not hold in embedding space. The model assumes the embedding trajectory is well approximated by a truncated Fourier series per coordinate. Fourier is the right basis for a periodic signal, but a contrastively trained encoder may vary <i>discontinuously</i> with surface state, in which case residuals are large everywhere and precision collapses. Note that prior embedding-based work fitted only the first principal component, or used per-month z -scoring, rather than a full harmonic model — plausibly for this reason.	Tier-1 physical indices are provably smooth and periodic, so the harmonic model is primary on Tier-1 and the embedding channel is corroborative rather than load-bearing. Decisive early experiment: measure harmonic fit R^2 in embedding space against R^2 in index space. One day of work, and it is the go/no-go on C1.
3	Conformal validity fails under residual shift. Exchangeability is an assumption, and a biased null set silently breaks the guarantee.	Lemma 11 bounds the null-contamination case: validity is preserved and the cost is paid in power, provided $\varepsilon < \alpha$. Assumption 4 is tested by Ljung–Box, not asserted. Report per-stratum coverage, not merely marginal. Treat under-coverage as a finding to be published, not hidden. Adaptive conformal maintains empirical coverage under drift.
4	Resolution/query mismatch. Sub-pixel targets (vehicles) are unanswerable from the mandated open sources (§3.5).	Stated openly in the architecture note. Aggregate-signature retrieval via optical/SAR discordance; honest low confidence; GSD-parameterised pipeline that consumes finer imagery if supplied.
5	Cross-sensor embedding misalignment. S1 and S2 embeddings are not comparable by default.	Per-sensor indices; sensor as a Mondrian stratum; CLOSP if staged. SAR used as a change and quality channel where a shared text space is unavailable.
6	Regional events defeat the control cohort. A district-wide flood floods the cohort too (Assumption 2).	Detected by the inlier-fraction diagnostic ρ and routed to a separate regional-event queue with the first-stage residual surfaced. A designed behaviour, not a failure.
7	Offline packaging failure on demo day. Missing basemap tiles, a CDN font, a runtime weight download.	Stage 0 concludes with a mandatory network-disabled rehearsal. This risk is eliminated by process, not by hope.
8	Scope. Six required capabilities plus a novel statistical layer is a large build.	The stage decomposition is explicitly ordered so that Stages 0–8 constitute a complete working system; Stages 9–11 are additive. The conformal layer (Stage 7) is thin — it is quantile arithmetic over an existing score — and is deliberately placed on top of a working detector rather than inside it.
9	Encoder choice sensitivity. No GFM dominates [12].	Encoder-agnostic architecture: every contribution operates on a residual series, whatever produced it. Encoder selection becomes an empirical ablation rather than a bet.

Table 11: Risk register.

Limitations we accept and state.

- Change localisation bottoms out at patch resolution (160 m). The morphology module (§5.7) recovers polygonal change objects at that granularity, which is finer than the 2.24 km tile and gives us exportable geometry, but it is still not a pixel-accurate mask; refining further would require a segmentation head that our embedding trajectories do not provide. We consider the trade justified — the problem statement asks for a ranked analyst queue with evidence, not a pixel-accurate map — but note the exposure: if the held-out change set is scored by pixel IoU rather than at candidate level, 160 m granularity will cost us.
- Assumption 1 (common shock) is a genuine assumption, not a theorem. We test it and report the test, but a strongly heterogeneous atmosphere within a single scene will degrade the estimator.
- The conformal guarantee is about false alarms under $\mathcal{H}_0$. It says nothing about power. A subtle change may be missed, and no amount of calibration fixes that; recall is governed by the encoder and the resolution, and we report it separately and honestly.
- Interferometric coherence — an excellent construction and clearance indicator, and cloud-immune — requires SLC products and an InSAR chain (SNAP/ISCE) that materially complicates offline staging. We scope it as an optional module and will state whether it was enabled in the evaluation.

11 Deliverables Traceability

Req.	Where satisfied	Evidence at evaluation
R1.1	Stage 5 (VLM channel), 8a, 9	Text query demo; nDCG on held-out queries
R1.2	Stage 8a, 8c	Image-to-image demo; Recall@ k
R1.3	§5.9 scoring; Stage 8c	Ranked list with scores
R1.4	Stage 8b (filtered ANN); Stage 9 META	Filter demo; selectivity/latency table
R1.5	§5.9.2; Stage 9 SPA	Relational query class in the retrieval breakdown
R2.1	Stage 7; §5.7 (morphology)	Change queue; P/R/ F_1 on held-out set; per-class metrics for appearance / disappearance / expansion / contraction
R2.2	§5.8; Stage 7	Conformal type sets; per-class metrics
R2.3	§5.6; Stage 7	(t^-, t^+, G) output; interval coverage and width
R3.1	Stages 2, 3, 6, 7 (C1, C2)	Confounder-stratified FAR table; ablation A2, A3
R3.2	Stage 2 ($q_i(t)$); Stage 7 (p, E)	Quality-weight propagation; confidence triple
R3.3	§5.4 (C3); BY-FDR	Calibration curve ; queue-level FDR
R4.1	Stage 8d (HDBSCAN over $\mathbf{v}^{\text{sem}} \parallel \mathbf{v}^{\text{phen}}$)	Cluster demo; purity
R4.2	Stage 8d (query-by-example, ζ slider)	Seed-site demo; precision@ k
R5.1	Stage 10a, 10b	Review-queue UI; time-series evidence panel
R5.2	Stage 10c (hash-chained audit)	Audit table; tamper-evidence demo
R5.3	Stage 10d	Rerank demo; calibration tightening after feedback
R5.4	Stage 10e	Export inspection; lineage round-trip
R6.1	Stage 8a (pgvector HNSW $\times 3$)	Index size and latency

Req.	Where satisfied	Evidence at evaluation
R6.2	Stage 11 (RLS, Sherman–Morrison, HNSW insert, e-process)	Live incremental-ingest demo with no rebuild
R6.3	Stage 0; whole stack	Network-disabled run
R6.4	Stage 1 (STAC); Stage 10e	Metadata round-trip
R6.5	Stage 1 (GDAL/rio-cogeo)	COG and GeoTIFF ingest demo
R7.1	Stage 0 sealing and rehearsal	Live air-gapped demonstration
R7.2	Table 7; <code>PROVENANCE.md</code>	Licence declarations and weight hashes
R7.3	Stage 0	Source manifest
R8.1–8.4	Repository; this document	Code, architecture note, procedures, provenance
R8.5	Stage 12	Auto-generated evaluation report

Table 12: Requirement-to-implementation traceability.

12 Conclusion

The problem statement asks for six capabilities. Systems that treat them as six subsystems will build six thresholds and spend their effort tuning them against a validation set they will not be scored on.

CHRONOS derives all six from a single modelling commitment: **a ground location is a trajectory, not a point**. From the harmonic decomposition of that trajectory we obtain a season-invariant retrieval vector, a phenological fingerprint for unsupervised discovery, and a residual series. From a second differencing against a phenologically matched control cohort we obtain a difference-in-differences estimand of real change with stated and testable identification assumptions. From conformal calibration of that estimand we obtain a distribution-free, finite-sample bound on the false-alarm rate; from an e-process over the conformal p-values we obtain an anytime-valid guarantee that survives unbounded incremental ingestion; and from Benjamini–Yekutieli we obtain a queue-level statement an analyst can act on. Relational language is routed to the operator that can answer it exactly rather than to one that is documented to answer it at chance.

The claim we are prepared to defend is narrow and specific. It is not that we have built a better encoder — we have not, and nobody should in this timeframe. It is that the field has been attacking a confounding problem with heuristics when it is in fact a well-posed statistical estimation problem with a known solution, and that stating it correctly yields not only better precision but a guarantee where previously there was only a tuned threshold. On a held-out evaluation containing deliberately adversarial no-change cases, the difference between a number one tuned and a number one guaranteed in advance is the difference between an engineering demonstration and a scientific result.

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